HASHICORP CERTIFIED · TERRAFORM ASSOCIATE 003

EXAM COMMAND CENTER

Reviews · 261 questions · 3 timed mocks · the shortest path to 70%. Assumes the Master series — this volume reviews and drills, it does not re-teach from zero.

Master Vol 1 ✓	Master Vol 2 ✓	Exam Command Center
Foundations	Production	← this book

CONTENTS

Command Map

116 SHEETS

Intelligence → arm each domain → drill → simulate → analyze → adjust. Every row jumps; every folio carries you back.

STUDY ROUTE

Intel → Arm → Drill → Simulate → Adjust

2 MIN

From intel (know the enemy: 57 questions, 63 seconds each, 40 correct passes) → arm each domain (D1 through D9, heaviest first if time is short: state 20% + config 18% ≈ 22 questions) → drill (90 domain questions) → simulate (three timed mocks) → analyze (keys) → adjust (tracker). The exam rewards recognition; the job rewards understanding — every explanation trains both.

> 🔑 THE 40-POINT PLAN
>
> Pass = 40/57. The highest-frequency archetypes alone cover it: state purposes + locking (D7), precedence + count/for_each (D8), workflow order (D6), workspaces (D3), modules sources (D5), Sentinel/TFC workspaces (D9). Master those six clusters and the remaining 17 points come from recognition.

▣Route by time available

You have	Do	Skip
One evening	Intel + D7 + D8 reviews + Mock 1	Drills; mocks 2–3
One weekend	All reviews + all drills + Mock 1 + Mock 2	Deep keys except misses
Two weeks	Everything in order; all three mocks timed; tracker green	Nothing

Know the Enemy: 57 Q · 60 Min · 70%

🎯 FACTS

The exam card (memorize)

Fact	Value
Questions	**57** · multiple choice + multi-select
Time	**60 minutes** → ~63 seconds per question
Pass	**70% ≈ 40 correct** — you may miss 17
Format	Proctored online · ~$70.50 · valid 2 years

▣Domain weights = your study budget

Domain	Wt	≈ Qs
D7 State	**20%**	~11
D8 Configuration	**18%**	~10
D9 Cloud/Enterprise	**13%**	~7
D3 Basics · D6 Workflow	9% each	~5+5
D2 Purpose · D4 CLI · D5 Modules	8% each	~5+5+4
D1 IaC concepts	7%	~4

▣The three-pass method (63-second economy)

PASS 1 · 30 min
answer everything, flag hard ones

→

PASS 2 · 20 min
flagged only, eliminate hard

→

PASS 3 · 10 min
review marked, never blank

Blank = guaranteed zero. A flagged guess beats an empty answer every time.

Morning-Of + Guaranteed-Pass Checklist

STRATEGY

The exam tests recognition under time pressure — not depth. Your edge is elimination speed: every distractor below is a documented misconception, and this book has trained each one.

▣Morning of (the boring list that works)

ID + quiet room + cleared desk ready before login

One mock warm-up (20 questions, untimed) — recognition primed

Arrive 15 min early; test the proctor software first

Water nearby; 60 minutes is a sprint, not a marathon

▣Guaranteed-pass checklist (40 points, highest frequency first)

#	Archetype	≈ pts	Armed at
1	State purposes + locking + force-unlock	6	D7
2	Precedence ladder + count/for_each + outputs	6	D8
3	Workflow order + plan modes + saved plans	5	D6
4	Workspaces vs directories + TFC workspaces	5	D3 + D9
5	Module sources + pins + registry types	4	D5
6	fmt/validate/console + lock file + TF_LOG	4	D4
7	IaC benefits + TF vs CFN/Ansible	5	D1 + D2
8	Sentinel + VCS runs + cost estimates	5	D9

Rows 1–8 ≈ 40 points = the pass line. Everything else is margin.

REVIEW 📖

Why "as Code" Beats Clicks

IaC manages infrastructure — networks, VMs, load balancers — in a **descriptive model**, versioned like application source. Same model → same environment, every time. The exam tests purpose and benefits, never syntax, here.

📌 THE ESSENCE — what the exam actually asks

Purpose: eliminate manual, unrepeatable provisioning; make environments reproducible

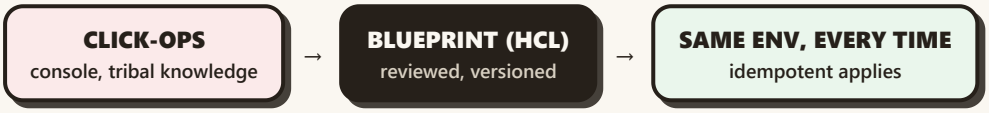
Declarative (declare WHAT — Terraform, CloudFormation) vs **imperative** (script the HOW — Ansible playbooks, bash)

Idempotence: same code applied twice = same result, second run changes nothing

Benefits: version control (git log = audit), repeatability, peer review (PRs), automation (CI), consistency (no snowflakes)

Golden rules: never hand-edit state · never hand-edit `.terraform/` · store code in VCS · treat infra like software

▣The mental model



The exam loves "which problem does IaC solve?" — answer: manual drift, unrepeatable envs, unaudited change.

⚠ TRAPS

"Imperative = better control" (exam wants declarative for infra) · "IaC means no manual steps ever" (backends/buckets are bootstrapped by hand first) · confusing idempotence with "runs fast".

Drill these → D1 drills (6 Q)

REVIEW 📖

What Terraform Is (and Isn't)

A tool for **building, changing, and versioning** infrastructure safely and efficiently — HashiCorp, 2014, Go, single binary, open source (MPL 2.0 line). Manages existing popular providers AND custom in-house solutions.

📌 THE ESSENCE

Three verbs: build · change · version — every exam stem about purpose uses them

Multi-provider universality: AWS + Cloudflare + GitHub in one apply; providers are plugins over RPC

Why HCL declarative: you state end-state; the graph + providers derive ordering and calls

vs CloudFormation: AWS-only, deep integration, no multi-cloud

vs Ansible: imperative config management, no state/plan for infrastructure

vs Pulumi: general-purpose languages instead of HCL — same declarative idea

💡 ELIMINATION KEY

Stems with "single cloud, deep AWS integration" → CloudFormation. "Configure OS/packages on existing hosts" → Ansible. "Version infrastructure across providers" → Terraform.

⚠ TRAPS

"Terraform is configuration management" (no — provisioning) · "Terraform only manages new infra" (adopts existing via import) · 📝 Modern note: 2023+ releases are BUSL; the MPL line continues as OpenTofu.

Drill these → D2 drills (8 Q)

Install, Load, Validate, Isolate

REVIEW 📖

Installing the binary, how configuration loads and merges, workspaces for env isolation, and the two inspection commands. Small domain, highly testable mechanics.

📌 THE ESSENCE

Install: single Go binary (package manager or direct download); upgrade = replace binary, then `init -upgrade`

Load order (last wins): defaults < `TF_VAR_*` < `terraform.tfvars` < `*.auto.tfvars` < `-var-file` < `-var`

Workspaces: `list` / `new` / `select` / `show` / `delete` — one backend, per-env state keys; `default` is implicit and undeletable

`terraform.workspace` in code = active workspace name (key into per-env maps)

`validate` = syntax + consistency, zero cloud · `fmt` = canonical style

Providers are plugins (RPC); declared in `required_providers`, installed by `init`, pinned by the lock file

▣The ladder, drawn (last wins, defaults lose)

Read right-to-left when debugging: the rightmost present layer is the value you get.

⚠ TRAPS

"`TF_VAR_*` beats files" (upside-down — files beat env) · "workspaces share state" (isolation is the point) · "validate checks the cloud" (never touches APIs).

Drill these → D3 drills (9 Q)

DOMAIN 4 · CLI · 8%

REVIEW 📖

Everything Outside the Core Loop

Format-on-save hygiene, credentials plumbing, version forensics, the console REPL, and log-driven debugging. The exam loves "which command" and "where is it stored" phrasing here.

📌 THE ESSENCE

`fmt` (+ `-check` gate, `-recursive`) — style machine; **`validate`** — syntax gate, no cloud

Credentials: `credentials.tfrc.json` (TFC tokens) + env vars (`AWS_*`, `TF_VAR_*`) — never committed secrets

`terraform login` — writes the TFC/TFE token to the credentials file

Provider versions in use: `.terraform.lock.hcl` (exact builds + hashes)

`terraform console` — REPL for functions/interpolation (`echo 'format(...)' | terraform console`)

`TF_LOG`: OFF, ERROR, WARN, INFO, DEBUG, **TRACE** (most verbose); `TF_LOG_PATH` captures to file

▣Verbosity ladder + credential map


```
$ TF_LOG=TRACE terraform apply 2> debug.log   # loudest → file
$ terraform console                           # REPL, no cloud
$ terraform login                             # token → credentials.tfrc.json
```

⚠ TRAPS

"`login` for AWS" (it's for Terraform Cloud) · "lock file is ignored" (it's committed) · "TRACE is quietest" (it's loudest).

Drill these → D4 drills (8 Q)

REVIEW 📖

Borrowed Code, Pinned Versions

Modules package infrastructure for reuse: the **root** module calls **child** modules by `source`, feeds them `inputs`, reads their `outputs`. Registry (public/private), git, local paths, S3 — sources vary, wiring never does.

📌 THE ESSENCE

Sources: local path · public registry · private registry · git / bitbucket · HTTP URLs · S3

Install: `init` downloads to `.terraform/modules` ; registry modules want `version` pins

Best practices: pin versions, standard structure (`main/variables/outputs`), no hardcoded providers inside

Wiring: root passes inputs → child exposes outputs → root (or stacks) consume `module.NAME.output`

Convention: `outputs.tf` is the module's public interface — reviewers read it first

```
module "network" {
  source  = "terraform-aws-modules/vpc/aws"   # registry
  version = "5.1.0"                            # ← ALWAYS pin registry modules
  name    = "prod"
  cidr    = "10.0.0.0/16"
}
# use: module.network.vpc_id
```

⚠ TRAPS

Unpinned registry source (floating majors) · provider blocks inside child modules (inherit from root) · reaching past outputs into internals.

Drill these → D5 drills (8 Q)

REVIEW 📖

Write → Init → Plan → Apply

The core loop with its modes and safety rails: plan flavors, saved-plan execution, auto-approve implications, and what interrupting an apply actually costs.

📌 THE ESSENCE

Order: write → `init` → `plan` → `apply` ; `plan` is read-only, always safe

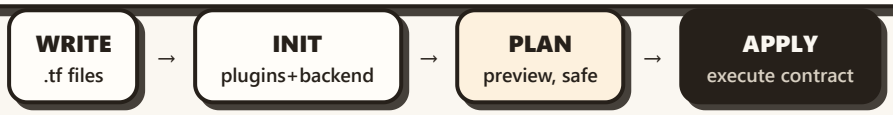
Modes: default (diff) · `-refresh-only` (drift preview/sync) · `-destroy` (teardown preview) · `-out=tfplan` (saved contract)

`apply tfplan` executes EXACTLY the reviewed plan; bare `apply` re-plans interactively

`-auto-approve`: CI behind gates only — hands type confirmation

Stopping apply mid-run: created resources stay, state records them, next apply continues — interruption orphans nothing, but review the partial state

Modules in workflow: `init` installs them; plans include them; pins freeze them



Dress rehearsal before opening night — reviewed == executed.

⚠ TRAPS

"Saved plan optional in CI" (it's the safety) · "interrupted apply rolls back" (nothing rolls back — state holds partial truth).

Drill these → D6 drills (9 Q)

The Ledger: Purposes, Remote, Locking

One-fifth of your exam. State maps config to reality, carries metadata, speeds plans, and serializes teams. Miss this domain and the math barely passes even if everything else is perfect.

📌 THE ESSENCE — purposes + remote + locking

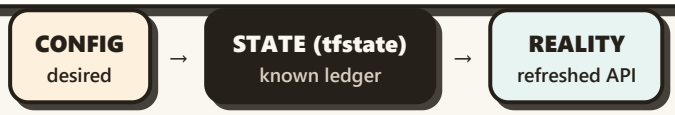
4 purposes: map config→real world · metadata store (IDs/IPs/ARNs) · performance cache · collaboration/locking point

Remote benefits: shared truth · **sensitive data off laptops** (still plaintext server-side!) · backend options (S3, TFC, Consul…) · locking

Locking: DynamoDB (or native `use_lockfile` 📝) serializes writers — one lock, one writer; stuck locks release by ID after proof of death

State is PLAINTEXT: secrets in state are readable — bucket SSE + tight IAM + Vault/SSM for true secrecy

Never hand-edit the JSON — every fix has a command (list/show/pull/push/rm/mv/import)



Delete the middle box → Terraform forgets everything → duplicates. That one sentence answers five exam questions.

⚠ TRAPS

"State encrypts secrets" (plaintext always) · "locking works on local state" (needs a shared backend) · `force-unlock` on a live holder (corruption).

Most-tested list → next page · Drills → D7 drills (18 Q)

Ranked: What the Exam Asks Most

#	Fact	Appears as...
1	State deleted → duplicates on next apply	scenario: "rm tfstate, apply, what happens?"
2	Secrets in state are plaintext	"which is true of sensitive values?"
3	Locking serializes; force-unlock by ID after proof	lock-error output diagnosis
4	`rm` forgets (lives) vs `destroy` deletes	"object survives but unmanaged — which cmd?"
5	`mv` renames tracking, zero rebuild	refactor-without-downtime stems
6	`import ADDR ID` adopts; config first	click-ops adoption ordering
7	`refresh-only` previews drift; bare refresh legacy	"show drift, change nothing"
8	Remote = shared + locked + off-laptop	"team of five, one state" scenarios

```
$ terraform state list   # what's managed
$ terraform state mv a b    # rename tracking   $ terraform import ADDR ID
$ terraform force-unlock ID # ghost lock, after proof
```

Drill these → D7 drills (18 Q) · Back to D7 p1

Address, Depend, Repeat, Vary

Second-heaviest domain: how resources are named and ordered, how copies multiply, how values flow in. Nearly every question here is answered by an address or a symbol.

📌 THE ESSENCE — addressing, deps, repetition, inputs

Addressing: `TYPE.NAME.attr` ; instances `[0]` (count) or `["key"]` (for_each); references create graph edges

Dependencies: implicit (references, always preferred) vs explicit (`depends_on` , invisible side-effects only)

count: integer copies, `count.index` ; shrinks kill the tail; 0/1 toggle idiom

for_each: set/map keys, `each.key/value` ; stable identity; never a bare list (wrap `toset`)

Variables: type + default + description + validation; full ladder (defaults LOSE, `-var` WINS)

Types: string/number/bool · list/set/map · object/tuple/any — sets unordered (for_each), lists indexed (count)

```
count = 3          →  web[0] web[1] web[2]      # position identity
for_each = toset(..) →  web["a"] web["b"]           # key identity
output "x" { value = aws_vpc.main.id }              # publish
locals { env = terraform.workspace }                # derive, don't repeat
```

⚠ TRAPS

for_each over a list (must be set/map) · `[0]` on empty (use one()) · depends_on as default (references first).

Most-tested list → next page · Drills → D8 drills (16 Q)

DOMAIN 8 · MOST-TESTED 🎯

Expressions, Splat, Pins

REVIEW 📖 2/2

#	Fact	Appears as...
1	Ladder: default < env < tfvars < auto < -var	"which value wins?" with 3+ layers
2	Count shrinks tail; keys hold identity	remove-middle predictions
3	~> = minor-flex, major-freeze	pin operator picks
4	Splat [*] lists all; for reshapes	"all IPs" vs "map of name→IP"
5	Ternary + one() for 0/1 resources	conditional reading
6	Dynamic blocks repeat nested blocks from vars	"N ingress rules from a list"
7	locals derive, resources declare	"where does this expression belong?"
8	Outputs publish; other stacks subscribe	cross-stack value stems

🎓 EXAM LENS

D8 rewards address fluency: cover the TYPE.NAME and every bracket in the question stem before reading the options — half the distractors die there.

```
{ for k, i in aws_instance.web : k => i.public_ip }  # map, not splat
count = var.on ? 1 : 0  +  one(x[*].arn)            # 0/1 idiom
version = "~> 5.0"  # 5.x yes, 6.0 never   dynamic "ingress" { for_each = var.rules … }
```

Drill these → D8 drills (16 Q) · Back to D8 p1

DOMAIN 9 · CLOUD/ENTERPRISE · 13%

REVIEW 📖

TFC, Sentinel, and the Other Workspaces

Terraform Cloud (SaaS) vs Enterprise (self-hosted): remote runs, VCS-driven plans, governance as code. Headline trap: **TFC workspaces ≠ CLI workspaces** — same word, different concept.

📌 THE ESSENCE

TFC = HashiCorp-hosted SaaS · **TFE** = self-hosted (your infra, your compliance boundary)

TFC workspaces bind config + variables + state + run history (NOT the CLI's state-key trick)

VCS integration: PR → speculative plan posted back; merge → apply (gated)

Sentinel policies: advisory (log) · **soft-mandatory** (override with permission) · **mandatory** (hard block)

Registries: public + private module AND provider registries; private = your org's blessed list

Runs: remote operations (plan/apply on TFC agents) · cost estimates on PRs · role-based access · free tier caps

The exam asks this distinction directly. Same word, different universe.

⚠ TRAPS

"Sentinel advisory blocks" (advisory only logs) · "TFC == TFE" (SaaS vs self-hosted) · agents run *remote* ops, not local ones.

Drill these → D9 drills (8 Q)

How the Mocks Work (Read First)

Three full simulations, 57 questions each, domains scattered like the real exam — **no domain badges on mock questions**, so every stem must be classified by YOU first. That's a trained skill: tag the domain before answering.

⏱ Timing discipline

- Write start time. 60 minutes, no pauses.
- Pass 1 (30m): answer all, flag hard.
- Pass 2 (20m): flagged only.
- Pass 3 (10m): review marked. Never blank.

📊 Score bands

- **<40:** back to reviews, not more mocks.
- **40–47:** pass — drill weak domains.
- **48–53:** strong, polish traps.
- **54+:** exam-ready. Book the date.

▣Answer grids + keys

Each mock ends with a self-marking grid (| Q | your letter |). Grids + hardest-20 deep explanations live in **mock keys**; every domain drill is fully explained at **drill keys**. Explanations teach mechanisms — a WHY that answers 5 future questions, not 1.

> 💡 **FIRST MOVE**
> Classify, then answer: state wording → D7 · brackets/pins → D8 · "team shares" → D7/D9 · Sentinel/VCS → D9 · "which command" → D4/D6.

D1 IaC + D2 openers

4 QUESTIONS

Q-D1-01 · Same config applied twice with no edits. The second plan reports…

A) No changes — Terraform skips planning

B) 1 to add, 0 to change, 0 to destroy

C) 0 to add, 1 to change, 0 to destroy

D) 0 to add, 0 to change, 0 to destroy

→ Explanation D1-01

Q-D1-02 · Which is the BEST definition of idempotence?

A) Code runs faster on repeat

B) Same code → same infra, re-apply changes nothing

C) Plans are always empty

D) State is optional

→ Explanation D1-02

Q-D1-03 · Your team reviews infra changes in PRs before they happen. Which IaC benefit is this?

A) Automation

B) Repeatability

C) Peer review

D) Speed

→ Explanation D1-03

Q-D1-04 · Declarative vs imperative: which pair is correct?

A) Terraform declares WHAT; scripted playbooks describe HOW

B) Terraform scripts HOW; playbooks declare WHAT

C) Both ignore state

D) Both are imperative

→ Explanation D1-04

EXAM LENS

D1 answers hide in definitions: idempotence, declarative, version-control benefits.

D1 end · D2 openers

4 QUESTIONS

Q-D1-05 · A junior hand-edits production in the console, then runs apply. What does Terraform do?

A) Detects drift via refresh and proposes to revert it

B) Ignores console edits forever

C) Deletes the state file

D) Locks the console out

→ Explanation D1-05

Q-D1-06 · Which file must NEVER be committed to version control?

A) main.tf

B) .terraform.lock.hcl

C) versions.tf

D) terraform.tfstate

→ Explanation D1-06

Q-D2-01 · Terraform is BEST described as...

A) A tool for building, changing, and versioning infrastructure

B) A configuration manager for OS packages

C) An AWS-only deployment service

D) A container orchestrator

→ Explanation D2-01

Q-D2-02 · Your stack spans AWS, Cloudflare, and GitHub in one apply. Which pillar explains why Terraform fits?

A) Speed and reliability

B) Platform-agnostic

C) Coding best practices

D) Massive community

→ Explanation D2-02

EXAM LENS

"Build, change, version" is the purpose answer; second API in the stem means platform-agnostic.

D2 purpose, set 2

4 QUESTIONS

Q-D2-03 · A question asks which tool versions infrastructure across clouds. The answer is...

A) Packer
B) Ansible
C) Terraform
D) Consul

→ Explanation D2-03

Q-D2-04 · "Configure Nginx and users on 200 existing servers." Best fit?

A) Terraform
B) Terraform Cloud
C) CloudFormation
D) Ansible

→ Explanation D2-04

Q-D2-05 · Which statement about the Terraform binary is TRUE?

A) It is a single Go executable with no runtime dependencies
B) It requires a Java runtime
C) It runs only on Linux
D) It embeds an AWS account

→ Explanation D2-05

Q-D2-06 · Why is HCL declarative an advantage for infrastructure?

A) It executes faster than binaries
B) You state end-state; ordering and API calls are derived
C) It needs no state
D) It skips planning

→ Explanation D2-06

EXAM LENS

Purpose stems answer with the three verbs; tool-fit stems match the layer.

Custom providers · install

4 QUESTIONS

Q-D2-07 · A provider for your in-house platform with a REST API. What do you know?

A) Impossible — only HashiCorp writes providers

B) Requires forking Terraform

C) Possible — providers are plugins; anyone can write one

D) Only via Sentinel

→ Explanation D2-07

Q-D2-08 · "AWS-only shop, deepest AWS integration wanted." You recommend…

A) Terraform, for multi-cloud

B) Packer, for AMIs

C) Ansible, for agents

D) CloudFormation, for native depth

→ Explanation D2-08

Q-D3-01 · Fresh laptop, first day. Which command sequence installs plugins for a new clone?

A) terraform init

B) terraform plan

C) terraform console

D) terraform fmt

→ Explanation D3-01

Q-D3-02 · Where do downloaded provider binaries live, and do they belong in git?

A) .terraform/ — committed

B) .terraform/ — ignored

C) terraform.tfstate — ignored

D) /usr/bin — committed

→ Explanation D3-02

EXAM LENS

Install = init; binaries = .terraform/ (ignored); lock = committed.

Precedence + workspaces

4 QUESTIONS

Q-D3-03 · Values set in defaults, TF_VAR_env, terraform.tfvars, and -var. Which wins?

A) defaults

B) TF_VAR_env

C) -var

D) terraform.tfvars

→ Explanation D3-03

Q-D3-04 · terraform workspace show prints…

A) the AWS region

B) all workspaces with resources

C) the backend bucket

D) the active workspace name only

→ Explanation D3-04

Q-D3-05 · Which workspace CANNOT be deleted?

A) default

B) staging

C) dev

D) prod

→ Explanation D3-05

Q-D3-06 · In code, terraform.workspace gives you…

A) the backend address

B) the active workspace name for lookups

C) a list of resources

D) the provider version

→ Explanation D3-06

EXAM LENS

Ladder ends at -var; default can never be deleted; workspace names key maps.

Pins · fmt gate

4 QUESTIONS

Q-D3-07 · terraform validate does what?

A) Refreshes state

B) Deploys to a sandbox

C) Checks syntax + consistency without touching the cloud

D) Formats files

→ Explanation D3-07

Q-D3-08 · Two teams need different instance sizes from one config. The primitive is…

A) a second repository

B) console edits

C) hardcoded locals

D) variables with per-team values

→ Explanation D3-08

Q-D3-09 · A provider release 6.0.0 must never install; 5.x bug-fixes may. Correct constraint?

A) version = "~> 5.0"

B) version = ">= 5.0"

C) version = "= 5.31.0"

D) no version key

→ Explanation D3-09

Q-D4-01 · CI must fail on unformatted HCL without rewriting. Command?

A) terraform fmt -check

B) terraform fmt

C) terraform validate

D) terraform plan

→ Explanation D4-01

EXAM LENS

Validate = syntax, no cloud. fmt -check gates; bare fmt rewrites.

Login · lock · console · logs

4 QUESTIONS

Q-D4-02 · Where does terraform login store the token?

A) credentials.tfrc.json

B) terraform.tfstate

C) .terraform.lock.hcl

D) An environment variable

→ Explanation D4-02

Q-D4-03 · Which file tells you the EXACT provider builds in use?

A) output.tf

B) versions.tf

C) terraform.tfstate

D) .terraform.lock.hcl

→ Explanation D4-03

Q-D4-04 · Test a format() expression without applying. Command?

A) terraform apply -target=null

B) echo expression | terraform console

C) terraform refresh

D) terraform show

→ Explanation D4-04

Q-D4-05 · Noisiest log level for a provider bug report?

A) WARN

B) OFF

C) INFO

D) TRACE

→ Explanation D4-05

EXAM LENS

login writes the credentials file; lock names builds; console tests; TRACE is loudest.

CI habits · first module

4 QUESTIONS

Q-D4-06 · Format-on-save in the editor plus what CI gate?

A) terraform apply

B) terraform fmt -check

C) terraform init

D) terraform destroy

→ Explanation D4-06

Q-D4-07 · AWS credentials for Terraform in CI come from…

A) hardcoded provider blocks

B) terraform.tfstate

C) env vars / OIDC role (never committed)

D) the lock file

→ Explanation D4-07

Q-D4-08 · Wrap terraform in Python for a pipeline. What must the wrapper preserve?

A) The .terraform directory in git

B) Colored output only

C) Exit codes and plan-file flow

D) Interactive prompts

→ Explanation D4-08

Q-D5-01 · A registry module must stay on 5.x. Correct source block?

A) local path with version key

B) source only

C) git URL with branch main

D) source + version = "5.1.0"

→ Explanation D5-01

EXAM LENS

Gates fail loudly; secrets never commit; registry modules pin versions.

Sources · install · wiring

4 QUESTIONS

Q-D5-02 · Valid module sources include...

A) only the public registry

B) local path, registry, git, HTTP/S3 URLs

C) only local paths

D) Docker Hub

→ Explanation D5-02

Q-D5-03 · Where do installed modules land after init?

A) The registry cache in /tmp

B) terraform.tfstate

C) .terraform/modules

D) Inside main.tf

→ Explanation D5-03

Q-D5-04 · Root vs child module: which is true?

A) Outputs flow downward only

B) Children call the root

C) Root calls children; children expose outputs the root consumes

D) There is no root module

→ Explanation D5-04

Q-D5-05 · A module block is missing version on a registry source. Risk?

A) Floating majors can break future runs

B) None — registry pins automatically

C) Init fails immediately

D) State is deleted

→ Explanation D5-05

EXAM LENS

Sources vary, wiring never does: inputs in, outputs out, version pinned.

Registries · the loop

4 QUESTIONS

Q-D5-06 · Private registry vs public registry: the difference is…

A) Language — private uses YAML

B) Cost — private is always paid

C) Speed — private skips init

D) Access scope — private hosts your org blessed modules

→ Explanation D5-06

Q-D5-07 · Provider blocks inside child modules: best practice?

A) Only for AWS

B) Avoid — inherit from root

C) Required in every child

D) Mandatory for pinning

→ Explanation D5-07

Q-D5-08 · How does a stack consume module.vpc.private_subnets[0]?

A) As a normal reference — outputs are values like any other

B) It cannot — outputs are display-only

C) Only in the console

D) Only via remote state

→ Explanation D5-08

Q-D6-01 · The core workflow in order is…

A) apply → plan → init → write

B) init → write → apply → plan

C) plan → write → init → apply

D) write → init → plan → apply

→ Explanation D6-01

EXAM LENS

Registries differ by scope; the loop never reorders: write, init, plan, apply.

Modes · contracts · interrupts

4 QUESTIONS

Q-D6-02 · "Show drift, change nothing." Command?

A) terraform refresh

B) terraform plan -refresh-only

C) terraform apply -auto-approve

D) terraform destroy

→ Explanation D6-02

Q-D6-03 · Why apply a saved plan file instead of bare apply?

A) Faster downloads

B) Skips init

C) Executes exactly the reviewed plan, no drift

D) Avoids state

→ Explanation D6-03

Q-D6-04 · -auto-approve in production by hand is…

A) Best practice

B) Acceptable Fridays

C) Dangerous — pipelines with gates only

D) Required for plan

→ Explanation D6-04

Q-D6-05 · Apply is interrupted halfway (Ctrl+C). What is true?

A) Created resources stay; state records them; next apply continues

B) Everything rolls back automatically

C) State is deleted

D) The backend is destroyed

→ Explanation D6-05

EXAM LENS

Refresh-only previews; saved plans bind; interrupts orphan nothing but finish nothing.

Destroy previews · care

4 QUESTIONS

Q-D6-06 · terraform plan -destroy -out=kill.plan does what?

A) Saves a teardown preview for later apply

B) Destroys immediately

C) Deletes the state

D) Removes the backend

→ Explanation D6-06

Q-D6-07 · A module version bumped upstream. Where does the workflow feel it?

A) init installs the new code; plan shows the resulting diff

B) Only in destroy

C) Nowhere — modules are static

D) State auto-upgrades

→ Explanation D6-07

Q-D6-08 · Plan shows -/+ on a database. Before confirming you…

A) Approve — replacements are routine

B) Edit state JSON by hand

C) Delete state and retry

D) Stop: replacement destroys data; snapshot/migrate first

→ Explanation D6-08

Q-D6-09 · Which plan mode syncs state toward reality deliberately?

A) Default plan

B) apply -refresh-only

C) plan -destroy

D) validate

→ Explanation D6-09

EXAM LENS

-/+ on stateful data stops the line. Destroy previews before destroy applies.

Purposes · remote · secrets · locks

4 QUESTIONS

Q-D7-01 · State purposes include all EXCEPT…

A) Compiling HCL to machine code

B) Storing metadata like IDs and ARNs

C) Mapping config to real objects

D) Speeding up plans via cache

→ Explanation D7-01

Q-D7-02 · A team of five shares one local state file. First fix?

A) Email the file weekly

B) Move to a remote backend with locking

C) Delete state after each apply

D) Disable refresh

→ Explanation D7-02

Q-D7-03 · Sensitive values in state are…

A) Encrypted by default

B) Hashed irreversibly

C) Plaintext — restrict bucket + IAM, use Vault/SSM

D) Never stored

→ Explanation D7-03

Q-D7-04 · Second apply starts while the first holds the lock. It…

A) Proceeds in parallel

B) Merges both writes

C) Deletes the lock

D) Fails with a lock error naming holder and ID

→ Explanation D7-04

EXAM LENS

D7 is 20%: purposes, plaintext secrets, one-writer locks.

Unlock · rm · mv · import

4 QUESTIONS

Q-D7-05 · Lock held by a dead CI runner. Correct recovery?

A) Re-run apply with -auto-approve until it passes

B) Delete the DynamoDB row by hand

C) Destroy and recreate the bucket

D) Confirm death, then force-unlock by ID, then prove health

→ Explanation D7-05

Q-D7-06 · terraform state rm aws_instance.old does what?

A) Stops managing it; the instance keeps running

B) Destroys the EC2

C) Renames it

D) Snapshots it first

→ Explanation D7-06

Q-D7-07 · Rename web → app with zero rebuild. Command?

A) terraform state mv aws_instance.web aws_instance.app

B) Edit state JSON in an editor

C) terraform destroy -target, then apply

D) terraform taint and apply

→ Explanation D7-07

Q-D7-08 · Adopt a hand-built VPC vpc-0ab1. First step?

A) terraform import immediately

B) Copy the state file

C) Delete it and let apply recreate

D) Write the matching resource block, then import, then plan clean

→ Explanation D7-08

EXAM LENS

rm abandons, mv renames, import adopts (config first).

Loss · pull/push · drift

4 QUESTIONS

Q-D7-09 · State file deleted, then apply. Result?

A) Clean no-op

B) Duplicate resources — Terraform forgot everything

C) Automatic restore from backend

D) Plan-only mode forever

→ Explanation D7-09

Q-D7-10 · terraform state pull > backup.json is used to…

A) Destroy faster

B) Upgrade providers

C) Snapshot raw state before risky surgery

D) Format code

→ Explanation D7-10

Q-D7-11 · state push is dangerous because it…

A) Formats JSON

B) Deletes the backend

C) Overwrites remote state with your local file

D) Rotates keys

→ Explanation D7-11

Q-D7-12 · Drift between config and reality is reconciled by…

A) Ignoring it — drift self-heals

B) Deleting .terraform

C) fmt -recursive

D) refresh during plan, then apply (or explicit refresh-only)

→ Explanation D7-12

EXAM LENS

Loss duplicates; pull backs up; push overwrites — direction matters.

Forks · hand-edits · outputs · tables

4 QUESTIONS

Q-D7-13 · Provider forked (new namespace). Existing state must follow. Command family?

A) init -upgrade

B) state replace-provider old/new

C) fmt -recursive

D) destroy then apply

→ Explanation D7-13

Q-D7-14 · Manually editing state JSON to "fix" a value is…

A) Best practice for speed

B) Forbidden — every fix has a command; hand edits corrupt ledgers

C) Required before import

D) Safe with backups

→ Explanation D7-14

Q-D7-15 · terraform show -json tfplan | jq .output_changes tells CI…

A) Formatting status

B) Provider download speed

C) What outputs will change, to gate deploys

D) The AWS bill

→ Explanation D7-15

Q-D7-16 · Lock table per env instead of one shared gives you…

A) Free state encryption

B) No locking at all

C) Faster plans

D) Parallel applies across envs at the cost of more tables to guard

→ Explanation D7-16

EXAM LENS

Forks need replace-provider; hands off the JSON; outputs gate pipelines.

Ephemeral · keys vs indexes

4 QUESTIONS

Q-D7-17 · terraform output -raw db_password in CI logs is a violation because…

A) Outputs are always encrypted

B) Sensitive display leaked to logs — mask output, pipe to a store

C) raw flag deletes state

D) JSON is required

→ Explanation D7-17

Q-D7-18 · ephemeral = true on an output means…

A) Never stored in state; same-apply read only

B) Encrypted at rest

C) Visible to everyone

D) Stored twice for safety

→ Explanation D7-18

Q-D8-01 · aws_subnet.public[0] vs aws_subnet.public["us-east-1a"] differ in…

A) Index identity vs key identity; only keys survive reordering

B) Nothing — identical

C) Speed only

D) Provider used

→ Explanation D8-01

Q-D8-02 · for_each over a bare list fails. Fix?

A) Use count instead always

B) Wrap toset(var.list) or build a key map

C) Delete the list

D) Add depends_on

→ Explanation D8-02

EXAM LENS

Ephemeral never persists; keys outlive positions.

Tails · one() · deps · dynamic

4 QUESTIONS

Q-D8-03 · count = 3 → count = 2 destroys…

A) web[0]

B) web[1]

C) web[2] — the tail

D) All three

→ Explanation D8-03

Q-D8-04 · Safe reader for a 0/1 counted resource?

A) x[0] directly

B) count.index

C) x[*][0]

D) one(x[*].attr)

→ Explanation D8-04

Q-D8-05 · Implicit vs explicit dependencies: prefer…

A) References (implicit); depends_on only for invisible side-effects

B) depends_on everywhere

C) Neither — random order is fine

D) depends_on instead of references

→ Explanation D8-05

Q-D8-06 · dynamic "ingress" with for_each over var.rules gives you…

A) N ingress blocks from one variable, zero HCL edits per rule

B) Faster plans

C) No state

D) Free subnets

→ Explanation D8-06

EXAM LENS

Tails die first; one() reads 0/1; references order; dynamics repeat.

Splat · required · sets · locals

4 QUESTIONS

Q-D8-07 · All public IPs as a list vs name→IP map: correct pair?

A) Maps are impossible

B) Both use [0]

C) [*].public_ip vs { for k,i : k => i.public_ip }

D) Splat returns a map

→ Explanation D8-07

Q-D8-08 · Variable with no default and nothing provided: plan…

A) Deletes state

B) Prompts (breaks CI) — provide via -var/env/file

C) Skips the resource

D) Uses empty string

→ Explanation D8-08

Q-D8-09 · set(string) vs list(string) for for_each: why sets?

A) No reason

B) Faster init

C) Sets allow duplicates

D) Stable unique keys; lists re-index on removal

→ Explanation D8-09

Q-D8-10 · locals { env = terraform.workspace } is an example of…

A) Hardcoding

B) Deriving, not repeating — single source per env

C) A provisioner

D) Backend config

→ Explanation D8-10

EXAM LENS

Splat lists, maps key; required vars prompt; sets key stably.

Conditionals · CIDR · merge

4 QUESTIONS

Q-D8-11 · Tribunal: ternary count = var.on ? 1 : 0 plus safe read gives…

A) A syntax error

B) Always-on resource

C) Conditional resource with one() output — null when off

D) Two resources

→ Explanation D8-11

Q-D8-12 · cidrsubnet("10.0.0.0/16", 8, 10) returns…

A) An error

B) 10.0.0.10/32

C) 10.10.0.0/16

D) 10.0.10.0/24

→ Explanation D8-12

Q-D8-13 · 5 subnets needed from a /16: newbits = ?

A) 3 (8 slots — enough)

B) 2 (4 slots — too few)

C) 5

D) 8

→ Explanation D8-13

Q-D8-14 · merge(var.tags, { Env = "prod" }) on key collision…

A) Errors

B) Right side wins

C) Left side wins

D) Both kept

→ Explanation D8-14

EXAM LENS

0/1 idiom, subnet math, last-wins merge.

Guards · TFC basics

4 QUESTIONS

Q-D8-15 · try(var.obj.attr, "fb") is for…

A) Encrypting values

B) Faster applies

C) Optional nested access without crashing

D) Deleting resources

→ Explanation D8-15

Q-D8-16 · validation { condition = contains([...], var.env) } runs…

A) At plan, rejecting bad input with your message

B) Never — decorative

C) Only in the console

D) After apply

→ Explanation D8-16

Q-D9-01 · TFC vs TFE: the difference is…

A) SaaS hosted by HashiCorp vs self-hosted in your boundary

B) Language used

C) Only the logo

D) TFE is older Terraform

→ Explanation D9-01

Q-D9-02 · A TFC workspace binds…

A) Only a state key suffix

B) Config + variables + state + run history

C) Just the CLI binary

D) Nothing — display only

→ Explanation D9-02

EXAM LENS

Guards fail fast; TFC workspaces are full environments, not labels.

VCS runs · Sentinel · registry

4 QUESTIONS

Q-D9-03 · PR opened → plan posted back → merge → apply. This is…

A) State mv

B) Local-exec

C) VCS-driven runs

D) Console mode

→ Explanation D9-03

Q-D9-04 · Policy blocks deploys missing cost tags, no override. Mode?

A) Advisory

B) Soft-mandatory

C) Disabled

D) Mandatory

→ Explanation D9-04

Q-D9-05 · Soft-mandatory vs mandatory Sentinel differs in…

A) Override possibility — permitted vs never

B) Speed

C) Language

D) Nothing

→ Explanation D9-05

Q-D9-06 · Private registry gives an org…

A) Faster CPUs

B) A blessed, versioned module/provider list

C) Free state

D) No init needed

→ Explanation D9-06

EXAM LENS

VCS drives runs; mandatory blocks; private registries bless.

Costs · agents + checkpoint

2 QUESTIONS

Q-D9-07 · Cost estimate on the PR comes from…

A) Local fmt

B) State mv

C) Remote runs analyzing the plan

D) Console

→ Explanation D9-07

Q-D9-08 · Agents in TFC/TFE run…

A) Local applies on laptops

B) Only fmt

C) Nothing — decorative

D) Remote operations on your infra

→ Explanation D9-08

EXAM LENS

Costs ride remote plans; agents bring runs to your network.

DRILL CHECKPOINT

90 drills banked. If D7/D8 misses cluster, re-arm those reviews before mocks — 38% of the exam lives there.

Questions 1–8

⏱ 60 MIN · 57 Q

🔑 Keys → M1-01 · M1-02 · M1-03 · M1-04 · M1-05 · M1-06 · M1-07 · M1-08

1. Your editor formats on save, but CI still needs a style gate. Which command fails the build on unformatted HCL?

A) terraform fmt -check

B) terraform fmt

C) terraform validate

D) terraform plan

2. What does terraform.tfstate primarily allow Terraform to do?

A) Compile HCL faster

B) Map configuration to real-world objects

C) Skip init

D) Encrypt variables

3. Two applies of unchanged config show 0/0/0. This demonstrates…

A) declarative syntax

B) remote backends

C) idempotence

D) peer review

4. Where are a managed VPC ID and its CIDR recorded after apply?

A) Nowhere — re-queried every second

B) Only in the console output

C) In versions.tf

D) In the state file as resource metadata

5. count = 3 is reduced to count = 2. Which instance is destroyed?

A) web[2]

B) web[0]

C) web[1]

D) All three

6. Which CANNOT be deleted?

A) A dev workspace

B) The default workspace

C) An empty workspace

D) A staging workspace

7. for_each requires its argument to be…

A) a list

B) a number

C) a set or map

D) a string

8. Terraform Cloud and Terraform Enterprise differ in…

A) state format

B) language: HCL vs YAML

C) CLI commands entirely

D) hosting: SaaS vs self-hosted

Questions 9–16

⏱ 60 MIN · 57 Q

🔑 Keys → M1-09 · M1-10 · M1-11 · M1-12 · M1-13 · M1-14 · M1-15 · M1-16

9. A policy must block deploys without cost tags, with no override. Sentinel mode?

A) Mandatory

B) Advisory

C) Soft-mandatory

D) Disabled

10. Token storage after terraform login?

A) terraform.tfstate

B) credentials.tfrc.json

C) .terraform.lock.hcl

D) An S3 bucket

11. Provider region B while default is region A. The resource needs...

A) a second Terraform install

B) depends_on

C) provider = aws.eu on the block

D) a second state file

12. TF_VAR_region, terraform.tfvars, and -var all set region. Winner?

A) defaults

B) TF_VAR_region

C) terraform.tfvars

D) -var

13. State file deleted before apply. Consequence?

A) Duplicate resources

B) Safe no-op

C) Auto-restore

D) Faster plans

14. "Single cloud, deepest native integration." Best fit?

A) Terraform

B) CloudFormation

C) Ansible

D) Packer

15. Registry module best practice for versions?

A) Omit version for latest

B) Branch main

C) Pin with version = "x.y.z"

D) No pin needed

16. PR opened, plan posted back automatically. This is...

A) Console mode

B) local-exec

C) State mv

D) VCS-driven runs

Questions 17–24

⏱ 60 MIN · 57 Q

🔑 Keys → M1-17 · M1-18 · M1-19 · M1-20 · M1-21 · M1-22 · M1-23 · M1-24

17. "Configure users and Nginx on 200 existing VMs." Best fit?

A) Ansible

B) CloudFormation

C) Packer

D) Terraform

18. "Error acquiring the state lock" appears. First step?

A) Destroy everything

B) Delete the DynamoDB row

C) Read the lock info: who, what op, since when

D) Disable locking

19. Correct core workflow order?

A) apply, plan, init, write

B) write, init, plan, apply

C) init, apply, plan, write

D) plan, write, apply, init

20. aws_subnet.public["us-east-1a"] addressing implies…

A) count indexing

B) A data source

C) No state

D) for_each keys

21. Same word, different concept in TFC vs CLI?

A) Workspace

B) Apply

C) Plan

D) State file

22. Safe reader for a possibly-empty counted list?

A) x[0]

B) one(x[*].attr)

C) count.index

D) x[*][0]

23. Saved plan file purpose?

A) Faster init

B) Skip state

C) Execute exactly the reviewed plan

D) Format code

24. Declarative means…

A) Script every API call

B) Manual ordering

C) No planning needed

D) State end-state; let the tool derive steps

Questions 25–32

🔑 Keys → M1-25 · M1-26 · M1-27 · M1-28 · M1-29 · M1-30 · M1-31 · M1-32

25. Remote state benefits include...

A) Shared truth, locking, secrets off laptops

B) No locking needed

C) No backend needed

D) Free encryption of code

26. Cost estimate on a PR is produced by...

A) Local fmt

B) Remote runs analyzing the plan

C) State mv

D) The console

27. References between resources create...

A) Nothing

B) Costs

C) Graph edges that order creation

D) Duplicates

28. terraform workspace show prints...

A) All resources

B) The backend type

C) Plan output

D) The active workspace name

29. Second run shows 0/0/0 after console tagging. Drift shows as...

A) A ~ change on the tagged resource

B) It cannot show

C) State deletion

D) A new workspace

30. TFC workspaces bind...

A) Only a key suffix

B) Config, variables, state, run history

C) Nothing

D) Local binaries

31. version = "~> 5.0" allows...

A) Only 5.0.0

B) Any version

C) 5.x, never 6.0

D) Only 6.x

32. terraform init does NOT...

A) Download providers

B) Configure the backend

C) Write the lock file

D) Create real infrastructure

Questions 33–40

🔑 Keys → M1-33 · M1-34 · M1-35 · M1-36 · M1-37 · M1-38 · M1-39 · M1-40

33. Soft-mandatory Sentinel differs from mandatory in...

A) Override possibility

B) Language

C) Speed

D) Nothing

34. Which workspace survives every delete attempt?

A) dev

B) default

C) staging

D) prod

35. Noisiest TF_LOG level?

A) OFF

B) INFO

C) TRACE

D) WARN

36. Sensitive values in state files are...

A) Encrypted by default

B) Hashed

C) Absent

D) Plaintext — restrict access

37. Child modules should receive values via...

A) Input variables from the root call

B) Console edits

C) Hardcoded locals

D) State surgery

38. Pulumi differs from Terraform mainly in...

A) No state

B) General-purpose languages vs HCL

C) AWS-only

D) No planning

39. Local path, registry, git, HTTP/S3 — these are...

A) Backend types

B) Log levels

C) Module source types

D) Workspace kinds

40. { for k,i : k => i.public_ip } builds...

A) A list

B) An error

C) A subnet

D) A name→IP map

Questions 41–48

⏱ 60 MIN · 57 Q

🔑 Keys → M1-41 · M1-42 · M1-43 · M1-44 · M1-45 · M1-46 · M1-47 · M1-48

41. terraform validate checks...
A) Syntax + consistency, no cloud
B) Real costs
C) State locks
D) DNS

42. Golden IaC rule about state files?
A) Commit them always
B) Never hand-edit; use state commands
C) Hand-edit freely
D) Email them

43. TF_VAR_tags for a map variable needs...
A) Bare braces
B) Double quotes only
C) Single-quoted HCL
D) A tfvars file only

44. terraform state pull is for...
A) Faster applies
B) Formatting
C) Provider upgrade
D) Raw JSON snapshot before surgery

45. Private registry gives...
A) Blessed org modules/providers
B) Faster CPUs
C) Free state
D) No init

46. terraform show -json tfplan | jq .output_changes gates...
A) Formatting
B) Deploys on output changes
C) Billing
D) DNS

47. TFC agents run...
A) Laptop applies
B) Nothing
C) Remote operations on your infra
D) Only fmt

48. merge(a, b) on colliding keys...
A) Errors
B) Left wins
C) Drops both
D) Right wins

Questions 49–56

🔑 Keys → M1-49 · M1-50 · M1-51 · M1-52 · M1-53 · M1-54 · M1-55 · M1-56

49. Interrupted apply (Ctrl+C) leaves...

A) Partial state; next apply continues

B) Full rollback

C) Deleted backend

D) No trace

50. validation blocks run...

A) After apply

B) At plan, rejecting bad input

C) In the console only

D) Never

51. -auto-approve belongs...

A) On laptops in prod

B) Everywhere

C) In gated pipelines only

D) Nowhere

52. terraform workspace new prod fails: exists. Meaning?

A) State corrupted

B) Init needed

C) Backend down

D) Name taken — select it instead

53. VCS integration posts plans...

A) Back on the PR

B) To email

C) To state

D) Nowhere

54. depends_on is for...

A) Every reference

B) Invisible side-effects only; references first

C) Deleting state

D) Formatting

55. terraform console is...

A) A backend

B) A REPL for expressions, no cloud

C) A deployment tool

D) A registry

56. force-unlock needs...

A) Nothing

B) Fmt first

C) A new bucket

D) The lock ID after proof the holder is dead

⏱ 60 MIN · 57 Q

Question 57 + answer sheet

57. Free-tier TFC caps push you to...

A) Paid tier or scoped usage

B) Share tokens

C) Commit state

D) Ignore limits

→ Key M1-57

▣Mock 1 answer sheet (mark, then check at keys)

Q	Yours	Q	Yours	Q	Yours
1	☐	2	☐	3	☐
4	☐	5	☐	6	☐
7	☐	8	☐	9	☐
10	☐	11	☐	12	☐
13	☐	14	☐	15	☐
16	☐	17	☐	18	☐
19	☐	20	☐	21	☐
22	☐	23	☐	24	☐
25	☐	26	☐	27	☐
28	☐	29	☐	30	☐
31	☐	32	☐	33	☐
34	☐	35	☐	36	☐
37	☐	38	☐	39	☐
40	☐	41	☐	42	☐
43	☐	44	☐	45	☐
46	☐	47	☐	48	☐
49	☐	50	☐	51	☐
52	☐	53	☐	54	☐
55	☐	56	☐	57	☐

Questions 1–8

⏱ 60 MIN · 57 Q

🔑 Keys → M2-01 · M2-02 · M2-03 · M2-04 · M2-05 · M2-06 · M2-07 · M2-08

1. State purposes do NOT include...
A) Compiling providers
B) Metadata storage
C) Mapping config to reality
D) Plan performance

2. Debug a provider with maximum logs to a file.
A) TF_LOG=OFF
B) TF_LOG=TRACE ... 2> debug.log
C) terraform fmt
D) rm .terraform

3. "Same environment from the same model, every time" is...
A) Snowflake servers
B) Manual clicking
C) Idempotence via IaC
D) Untracked change

4. TFC free tier limits push heavy teams to...
A) Local backends only
B) Committing state
C) Disabling plans
D) Paid tier or scoped usage

5. aws_instance.web["blue"] implies...
A) for_each keys
B) count
C) No backend
D) A provisioner

6. Pulumi vs Terraform: core difference?
A) General-purpose languages vs HCL
B) No state in Pulumi
C) Pulumi is AWS-only
D) Terraform has no plan

7. Registry child module needs a bug-fix version safely. Pin?
A) Branch latest
B) No pin
C) version = "~> 2.0"
D) version = ">= 1.0"

8. TF_VAR_count="three" for a number var: result?
A) Auto-corrects to 3
B) Destroys state
C) Ignored silently
D) Type error at plan

Questions 9–16

⏱ 60 MIN · 57 Q

🔑 Keys → M2-09 · M2-10 · M2-11 · M2-12 · M2-13 · M2-14 · M2-15 · M2-16

9. Write → init → plan → apply. Skipping plan risks…
A) Faster runs only
B) Unreviewed changes executing blind
C) Nothing
D) Better state

10. Golden rule: infra change without a PR is…
A) Unreviewed — violates IaC peer review
B) Faster
C) Required
D) Encrypted

11. DynamoDB lock table deleted. Applies now…
A) Run faster
B) Encrypt state
C) Race without serialization — restore it
D) Skip init

12. terraform output -json | jq .vpc_id.value is for…
A) Formatting
B) Init
C) Locking
D) CI/script consumption of exports

13. Speculative plan on a PR comes from…
A) VCS-driven runs
B) local-exec
C) fmt
D) console

14. Lock error names ci-runner-9, job red for an hour. Move?
A) Wait forever
B) Confirm death, force-unlock ID, prove health
C) Delete bucket
D) Reboot laptop

15. Public registry vs private module registry: use private for…
A) Avoiding init
B) Faster downloads only
C) Blessed org versions
D) Public code only

16. values(aws_subnet.public)[*].id returns…
A) One ID
B) A map
C) Nothing
D) All IDs as a list

Questions 17–24

🔑 Keys → M2-17 · M2-18 · M2-19 · M2-20 · M2-21 · M2-22 · M2-23 · M2-24

17. terraform workspace list shows...
A) All workspaces, * marking active
B) Only the current workspace
C) Resource counts
D) AWS costs

18. Advisory Sentinel policy violation...
A) Blocks
B) Logs only, run continues
C) Deletes state
D) Bills extra

19. lookup(var.amis, var.region, "ami-fallback") does...
A) Crashes on miss
B) Deletes AMIs
C) Returns fallback on miss
D) Ignores region

20. Manual console SG edit appears in next plan as...
A) Nothing
B) An error, always
C) A new workspace
D) A ~ change to reconcile

21. terraform apply tfplan guarantees...
A) Reviewed == executed
B) No state
C) No backend
D) Faster init

22. terraform state show is used to...
A) Delete resources
B) Display one resource attributes
C) Format code
D) Create workspaces

23. Mandatory policy with missing tags...
A) Warns only
B) Applies anyway
C) Hard-blocks the run
D) Deletes the policy

24. -refresh-only mode is for...
A) Destroying
B) Init
C) Formatting
D) Previewing/syncing drift explicitly

Questions 25–32

⏱ 60 MIN · 57 Q

🔑 Keys → M2-25 · M2-26 · M2-27 · M2-28 · M2-29 · M2-30 · M2-31 · M2-32

25. apply -refresh-only writes...
A) Synced state toward reality
B) Nothing
C) A new backend
D) Plan to trash

26. Remote operations (TFC agents) mean...
A) No history kept
B) Plans/applies run off-laptop with history
C) Local-only runs
D) No variables

27. New workspace starts with...
A) Copied prod state
B) All resources
C) Empty state — apply to fill
D) An error

28. coalesce(var.a, var.b, "fb") returns...
A) Always "fb"
B) Nothing
C) A list
D) First non-null/non-empty

29. terraform fmt -recursive covers...
A) Subfolders too
B) One file
C) State
D) The registry

30. Module outputs flow...
A) Downward only
B) Upward to root/consumers
C) Nowhere
D) To S3 directly

31. Root module is...
A) The state file
B) A registry only
C) The entrypoint config that calls children
D) The lock file

32. -destroy plan mode previews...
A) Creation
B) Init
C) Formatting
D) Teardown for later apply

Questions 33–40

⏱ 60 MIN · 57 Q

🔑 Keys → M2-33 · M2-34 · M2-35 · M2-36 · M2-37 · M2-38 · M2-39 · M2-40

33. *.auto.tfvars vs terraform.tfvars differ in...
A) Layered lexical overlays vs single file
B) Nothing
C) Only comments
D) Speed

34. echo expression | terraform console tests...
A) Full applies
B) Functions/interpolation with no cloud
C) State locks
D) Billing

35. Ansible manages infra state like Terraform? No — because...
A) It is newer
B) It is faster
C) It has no state/plan for infra
D) It costs more

36. Dynamic blocks repeat...
A) Providers
B) Workspaces
C) Backends
D) Nested blocks from variables

37. TFC cost estimates appear...
A) On PRs from remote plans
B) In state
C) In fmt output
D) Never

38. Bare apply in prod is replaced by...
A) destroy
B) apply tfplan after review
C) rm state
D) console edits

39. Soft-mandatory violation with override rights...
A) Blocks always
B) Deletes policy
C) Proceeds with permission
D) Ignores code

40. terraform state list shows...
A) Nothing useful
B) AWS bills
C) Provider source
D) Every managed address

Questions 41–48

⏱ 60 MIN · 57 Q

🔑 Keys → M2-41 · M2-42 · M2-43 · M2-44 · M2-45 · M2-46 · M2-47 · M2-48

41. Module pin missing on registry source risks...
A) Floating majors breaking runs
B) Nothing
C) Instant failure
D) State loss

42. terraform.tfvars committed with secrets. Fix?
A) Nothing — normal
B) Gitignore + example template + secret store
C) Delete git
D) Encrypt filenames

43. depends_on belongs where?
A) Everywhere
B) Nowhere
C) Invisible side-effects only
D) In outputs

44. api.example.com CNAME via Cloudflare provider in one apply shows...
A) Lock failure
B) A bug
C) Drift
D) Multi-provider power

45. terraform validate needs...
A) Cloud credentials
B) No cloud — syntax only
C) A backend
D) State

46. Backend block with var.* fails because...
A) Vars are broken
B) Backends take literals/-backend-config only
C) S3 is down
D) Init is slow

47. for_each = toset(var.list) converts...
A) Map to string
B) Number to bool
C) List to stable set
D) Nothing

48. terraform show reads...
A) DNS
B) The future
C) Billing
D) State or plan files human/JSON

Questions 49–56

⏱ 60 MIN · 57 Q

🔑 Keys → M2-49 · M2-50 · M2-51 · M2-52 · M2-53 · M2-54 · M2-55 · M2-56

49. TRACE vs INFO: TRACE is...
A) Loudest, full RPC detail
B) Off
C) Quieter
D) Errors only

50. Orphaned EC2 found running, untracked. First command?
A) destroy
B) terraform state list to confirm absence, then import flow
C) rm -rf
D) fmt

51. TFC run history lives...
A) Nowhere
B) Locally only
C) In the workspace (config+vars+state+runs)
D) In git

52. Module output consumed as...
A) Via email
B) Only via console
C) Never
D) module.NAME.output like any value

53. "Same model, same env" is IaC...
A) Risk
B) Slowness
C) Cost
D) Repeatability

54. Workspaces isolate...
A) Nothing
B) State per env under one backend
C) Providers
D) Binaries

55. -out=tfplan then apply tfplan gives...
A) Faster init
B) No state
C) Reviewed == executed
D) No backend

56. one(null_resource.none[*].id) returns...
A) An error
B) A list
C) Zero
D) null

⏱ 60 MIN · 57 Q

Question 57 + answer sheet

57. Old tutorial says "terraform refresh". Modern equivalent?

A) terraform plan -refresh-only

B) Nothing — removed

C) terraform destroy

D) terraform fmt

→ Key M2-57

▣Mock 2 answer sheet (mark, then check at keys)

Q	Yours	Q	Yours	Q	Yours
1	☐	2	☐	3	☐
4	☐	5	☐	6	☐
7	☐	8	☐	9	☐
10	☐	11	☐	12	☐
13	☐	14	☐	15	☐
16	☐	17	☐	18	☐
19	☐	20	☐	21	☐
22	☐	23	☐	24	☐
25	☐	26	☐	27	☐
28	☐	29	☐	30	☐
31	☐	32	☐	33	☐
34	☐	35	☐	36	☐
37	☐	38	☐	39	☐
40	☐	41	☐	42	☐
43	☐	44	☐	45	☐
46	☐	47	☐	48	☐
49	☐	50	☐	51	☐
52	☐	53	☐	54	☐
55	☐	56	☐	57	☐

Questions 1–8

⏱ 60 MIN · 57 Q

🔑 Keys → M3-01 · M3-02 · M3-03 · M3-04 · M3-05 · M3-06 · M3-07 · M3-08

1. "Build, change, version infrastructure" describes...
A) Terraform
B) Packer
C) Ansible
D) Git

2. Private provider registry is for...
A) Public mirrors only
B) Blessed org providers
C) Skipping init
D) Billing

3. terraform init -upgrade does what?
A) Destroys state
B) Formats code
C) Re-resolves newest allowed providers, rewrites lock
D) Creates workspaces

4. Saved destroy plan via...
A) login
B) fmt
C) console
D) plan -destroy -out=kill.plan

5. Module source pinned with version is...
A) Best practice for registry modules
B) Optional
C) Forbidden
D) Slow

6. Remote state benefits exclude...
A) Shared truth
B) Local-only secrecy
C) Locking
D) Off-laptop secrets

7. toset(var.list) is needed because for_each takes...
A) Lists happily
B) Numbers
C) Sets or maps only
D) Nothing

8. Ghost lock recovery starts with...
A) Reboot
B) Deleting the table
C) Destroy
D) Reading lock info

Questions 9–16

🔑 Keys → M3-09 · M3-10 · M3-11 · M3-12 · M3-13 · M3-14 · M3-15 · M3-16

9. Provider credentials belong in…

A) Env vars / SSO, never git

B) Committed blocks

C) State outputs

D) Lock file

10. "OS users and packages on live hosts" →

A) Terraform

B) Ansible

C) CloudFormation

D) TFC

11. Conditional resource idiom?

A) No conditionals exist

B) count = always

C) count = var.on ? 1 : 0

D) Destroy first

12. Peer review of infra happens via…

A) Console clicks

B) State edits

C) Phone calls

D) VCS PRs on HCL

13. values(map)[*].id collects…

A) All values IDs as a list

B) Keys

C) Nothing

D) One string

14. Agents run plans…

A) On laptops

B) Remotely with history

C) Never

D) In fmt

15. Explicit credentials in a committed block risk…

A) Nothing

B) Faster plans

C) Scraped secrets in minutes

D) Better state

16. Stale local state vs changed reality reconciles via…

A) New workspace

B) Deleting config

C) Fmt

D) Refresh during plan

Questions 17–24

🔑 Keys → M3-17 · M3-18 · M3-19 · M3-20 · M3-21 · M3-22 · M3-23 · M3-24

17. Speculative PR plans come from...
A) VCS-driven runs
B) Cron fmt
C) Local state
D) Email hooks

18. terraform state show vpc reads...
A) All bills
B) One resource attributes
C) Provider source
D) Nothing

19. merge() collision winner is...
A) Left
B) Error
C) Right
D) Neither

20. terraform.tfvars vs *.auto.tfvars differ in...
A) Nothing
B) Color
C) Speed
D) Single file vs layered lexical overlays

21. Root passes inputs to children via...
A) Module call arguments
B) State surgery
C) Console
D) Email

22. Console drift on tags appears as...
A) No output
B) ~ change
C) New backend
D) Crash

23. coalesce picks...
A) Last null
B) All values
C) First non-null
D) None

24. *.tfvars.example committed while real tfvars ignored because...
A) Speed
B) State needs it
C) Required by law
D) Template shared, secrets stay local

Questions 25–32

⏱ 60 MIN · 57 Q

🔑 Keys → M3-25 · M3-26 · M3-27 · M3-28 · M3-29 · M3-30 · M3-31 · M3-32

25. Interrupted mid-apply: next step?

A) Review partial state, re-apply to continue

B) Delete everything

C) New backend

D) Fmt

26. "No manual snowflakes" is IaC...

A) Cost

B) Consistency

C) Slowness

D) Risk

27. Go binary, no runtime describes...

A) The provider only

B) State files

C) The Terraform CLI distribution

D) Sentinel

28. Same config, two applies, both green 0/0/0. Property?

A) Import

B) Drift

C) Taint

D) Idempotence

29. try() guards...

A) Optional access without crashing

B) Billing

C) Locks

D) DNS

30. Graph edges derive from...

A) File order

B) References

C) Comments

D) Filenames

31. console tests...

A) Locks

B) Full deploys

C) Expressions with no cloud

D) Bills

32. TFC workspace binds...

A) Binaries

B) A key suffix only

C) Nothing

D) Config, vars, state, runs

Questions 33–40

⏱ 60 MIN · 57 Q

🔑 Keys → M3-33 · M3-34 · M3-35 · M3-36 · M3-37 · M3-38 · M3-39 · M3-40

33. Multi-cloud in one apply is possible because...
A) Providers are plugins over RPC
B) One giant binary
C) No state
D) YAML

34. Cost control on PRs via...
A) Fmt
B) Remote-run estimates
C) State rm
D) Taint

35. Splat [*] over instances gives...
A) A map
B) An error
C) A list of the attribute
D) A backend

36. Ghost lock: first read...
A) Fmt
B) Nothing
C) Billing
D) Lock info: who/what/since

37. Log verbosity for deep debugging?
A) TRACE
B) ERROR
C) OFF
D) WARN

38. apply -refresh-only syncs...
A) Code style
B) State toward reality
C) Providers
D) Regions

39. Advisory policy breach...
A) Blocks
B) Deletes
C) Logs, run continues
D) Bills

40. Partial apply then re-apply...
A) Rolls back
B) Forks code
C) Wipes backend
D) Continues from recorded state

Questions 41–48

⏱ 60 MIN · 57 Q

🔑 Keys → M3-41 · M3-42 · M3-43 · M3-44 · M3-45 · M3-46 · M3-47 · M3-48

41. "Golden image with Nginx baked" is built by...
A) Packer
B) Terraform provisioners
C) Console
D) State mv

42. for_each over ["a","b"] fails: fix?
A) Delete list
B) toset() wrap
C) depends_on
D) Count only

43. TF_VAR for objects needs...
A) Bare words
B) Double binaries
C) Single-quoted HCL
D) No quoting

44. Child output consumed as...
A) Never
B) Console only
C) Email
D) module.NAME.output

45. terraform.tfvars purpose?
A) Auto-loaded variable values
B) Backend only
C) Locking
D) Billing

46. Lock table per env buys...
A) Nothing
B) Parallel applies, more tables
C) No init
D) Free state

47. validation blocks run...
A) Never
B) After destroy
C) At plan
D) In git

48. -destroy plan mode is for...
A) Login
B) Creation
C) Fmt
D) Teardown preview

Questions 49–56

🔑 Keys → M3-49 · M3-50 · M3-51 · M3-52 · M3-53 · M3-54 · M3-55 · M3-56

49. New workspace state starts...
A) Empty — apply fills it
B) With prod data
C) With errors
D) Locked forever

50. TFC workspace binds...
A) Key suffix only
B) Config, vars, state, runs
C) Nothing
D) Binaries

51. Login writes token to...
A) State
B) Lock file
C) credentials.tfrc.json
D) tfvars

52. locals derive values to...
A) Duplicate code
B) Slow plans
C) Skip state
D) Avoid repetition, single source

53. terraform output shows...
A) Exports after apply
B) Billing
C) Locks
D) Nothing

54. Pin registry modules because...
A) Speed
B) Floating majors break runs
C) State needs it
D) Fmt does

55. Soft-mandatory with rights...
A) Blocks
B) Deletes
C) Proceeds permitted
D) Ignores

56. -auto-approve belongs...
A) Laptops in prod
B) Never anywhere
C) Always
D) Gated pipelines only

Question 57 + answer sheet

57. S3 backend state with secrets is…

A) Plaintext — restrict + Vault/SSM

B) Encrypted by Terraform

C) Absent

D) Hashed

→ Key M3-57

▣Mock 3 answer sheet (mark, then check at keys)

Q	Yours	Q	Yours	Q	Yours
1	☐	2	☐	3	☐
4	☐	5	☐	6	☐
7	☐	8	☐	9	☐
10	☐	11	☐	12	☐
13	☐	14	☐	15	☐
16	☐	17	☐	18	☐
19	☐	20	☐	21	☐
22	☐	23	☐	24	☐
25	☐	26	☐	27	☐
28	☐	29	☐	30	☐
31	☐	32	☐	33	☐
34	☐	35	☐	36	☐
37	☐	38	☐	39	☐
40	☐	41	☐	42	☐
43	☐	44	☐	45	☐
46	☐	47	☐	48	☐
49	☐	50	☐	51	☐
52	☐	53	☐	54	☐
55	☐	56	☐	57	☐

✅ D1-01 · D1-02 · D1-03

FULL WHY

Q-D1-01 ✅ **D —** 0 to add, 0 to change, 0 to destroy — plus the "No changes" line.
🧠 WHY — Refresh re-read reality, compare found zero drift: desired == known == real. An empty diff is the healthy steady state, not a skipped run. Idempotence means the second apply is a no-op by design.
❌ A/B/C — any nonzero count claims changes that do not exist
❌ D — "skips planning" mistakes the outcome for the process — Terraform planned fully, then did nothing
🎓 EXAM — Same code twice → 0/0/0. Any nonzero number needs a cause (edits, drift, taint).

Q-D1-02 ✅ **B —** Idempotence = same code → same infra, re-apply changes nothing.
🧠 WHY — It is a property of the RESULT across runs, not of speed, plans, or state files. "Faster on repeat" describes caching; "plans always empty" ignores first applies and real changes.
❌ A — speed is irrelevant — a slow idempotent apply is still idempotent
❌ C — first applies and drifted states produce full diffs
❌ D — state is the mechanism that MAKES idempotence possible
🎓 EXAM — When a stem says "twice, unchanged", answer idempotence before finishing the sentence.

Q-D1-03 ✅ **C —** Reviewing infra in PRs before it happens = peer review.
🧠 WHY — IaC puts infrastructure into reviewable text, so the same PR machinery that guards app code guards prod. Automation runs it; repeatability stabilizes it; neither involves human judgment.
❌ A — automation executes — it does not judge
❌ B — repeatability is about identical outcomes, not approval
❌ D — speed is never the point of a review gate
🎓 EXAM — PR + infra + "before" → peer review. "Same every time" → repeatability.

✅ D1-04 · D1-05 · D1-06

FULL WHY

Q-D1-04 ✅ **A —** Declarative states WHAT (Terraform); imperative scripts HOW (playbooks).
🧠 WHY — Terraform describes end-state and derives ordering plus API calls; Ansible-style playbooks list steps to execute. The exam pairs the paradigm with the tool without exception.
❌ B — exact inversion — the exam's favorite trap shape
❌ C — playbooks are the imperative half of the pair
❌ D — state is central to Terraform; playbooks simply lack it
🎓 EXAM — WHAT=HCL/Terraform, HOW=steps/playbooks. Inverted options are always wrong.

Q-D1-05 ✅ **A —** Console tag appears as a ~ change; refresh caught the drift.
🧠 WHY — Refresh imports REAL into the diff, so hand edits surface as update-in-place lines against code. The plan is the drift report nobody asked anyone to write.
❌ B — nothing is invisible to refresh — that is its entire job
❌ C — state is read, never deleted, by planning
❌ D — no console can lock Terraform out of reading APIs
🎓 EXAM — Hand edit + next plan = ~ line. Resolve by code (apply) or console (revert).

Q-D1-06 ✅ **D —** Never commit terraform.tfstate.
🧠 WHY — State holds secrets in plaintext plus the keys to everything. versions.tf and the lock file SHOULD commit (reproducibility); main.tf is the whole point of version control.
❌ A — main.tf is exactly what VCS is for
❌ C — versions.tf pins belong in git
❌ D — lock file guarantees identical providers
🎓 EXAM — State + secrets + git = breach. tfstate, tfvars values, and keys stay local.

✅ D2-01 · D2-02 · D2-03

FULL WHY

Q-D2-01 ✅ **A —** Build, change, version — the three verbs.
🧠 WHY — Terraform exists to take infrastructure through its whole lifecycle under version control. "Configuration manager" describes Ansible; "AWS-only" describes CloudFormation; "orchestrator" describes Nomad/K8s.
❌ B — OS packages = config management, wrong layer
❌ C — AWS-only contradicts multi-provider universality
❌ D — containers are someone else's job
🎓 EXAM — Purpose stems answer with the three verbs, every time.

Q-D2-02 ✅ **B —** Platform-agnostic wins the moment a second API enters.
🧠 WHY — AWS + Cloudflare + GitHub in one apply is impossible for CloudFormation and pointless to split across tools. Community, speed, and practices are all true of Terraform and all miss the question asked.
❌ A — speed never answers a scope question
❌ C — practices govern HOW, not WHERE
❌ D — community is real and irrelevant here
🎓 EXAM — Second API in the stem → platform-agnostic. Always.

Q-D2-03 ✅ **C —** Versions-infrastructure-across-clouds = Terraform.
🧠 WHY — Packer bakes images, Ansible configures hosts, Consul discovers services. Only one tool on the list versions heterogeneous infrastructure.
❌ A — Packer builds AMIs; it manages no live infra
❌ B — Ansible has no state/plan for infrastructure
❌ D — Consul is service discovery, not provisioning
🎓 EXAM — Tool-fit questions eliminate by layer: provision, configure, bake, discover.

✅ D2-04 · D2-05 · D2-06

FULL WHY

Q-D2-04 ✅ **D —** 200 existing servers needing users and Nginx = Ansible.
🧠 WHY — The work is OS-level configuration of long-lived hosts — the config-management layer, where idempotent modules beat provisioning tools. Terraform provisions; it does not babysit fleets.
❌ A — Terraform would need a provider per package manager — absurd
❌ B — CloudFormation is AWS-only AND provisioning-layer
❌ C — TFC runs Terraform; wrong tool, fancier venue
🎓 EXAM — "Existing hosts + packages/users" → Ansible. "New infra + lifecycle" → Terraform.

Q-D2-05 ✅ **A —** Single Go executable, zero runtime dependencies.
🧠 WHY — One static binary is the whole distribution story: download, run, no interpreter, no agents. Java runtimes, Linux-only limits, and embedded AWS accounts are all fictions.
❌ B — Go compiles statically — no JVM anywhere
❌ C — all major OSes ship builds
❌ D — it calls AWS APIs; it contains no account
🎓 EXAM — Distribution questions test "single binary, Go, no runtime" — one fused fact.

Q-D2-06 ✅ **B —** Declare end-state; ordering and API calls derive.
🧠 WHY — HCL says WHAT the world should look like; the graph derives creation order and providers translate intent into calls. Nothing executes faster, stateless, or plan-less.
❌ A — speed is not a semantic property
❌ C — state is what makes declaration work
❌ D — planning is the derivation step itself
🎓 EXAM — Declarative = describe the destination, let the machine route.

✅ D2-07 · D2-08 · D3-01

FULL WHY

Q-D2-07 ✅ **C —** Anyone can write a provider — plugins over RPC.
🧠 WHY — Providers are separate processes behind a defined interface; HashiCorp owns some, the community and vendors own hundreds more, and your team can own one for internal APIs.
❌ A — the registry alone disproves exclusivity
❌ B — no fork needed — that is what the plugin boundary is for
❌ D — Sentinel governs policy, not APIs
🎓 EXAM — In-house API + "can we manage it" → yes, custom provider.

Q-D2-08 ✅ **D —** Deepest AWS-native integration = CloudFormation.
🧠 WHY — Single-cloud depth is CloudFormation's entire value proposition: free, native, StackSets, auto-rollback — at the price of every other cloud. The stem asked for depth, not breadth.
❌ A — multi-cloud is the opposite of the ask
❌ B — AMIs don't integrate CloudFormation deeper
❌ C — agents solve configuration, not nativeness
🎓 EXAM — Ask what the stem prices: depth → CFN, breadth → Terraform, fleet config → Ansible.

Q-D3-01 ✅ **A —** New clone needs terraform init first.
🧠 WHY — No plugins, no modules, no backend — nothing works until init downloads and wires them. Plan, console, and fmt all assume init already ran.
❌ B — plan needs providers to refresh anything
❌ C — console evaluates against installed providers
❌ D — fmt works but proves nothing about readiness
🎓 EXAM — Empty dir + "first" → init. Always.

FULL WHY

✅ D3-02 · D3-03 · D3-04

Q-D3-02 ✅ **B —** .terraform/ holds binaries and is ignored.
🧠 WHY — Downloaded plugins are re-fetchable weight: identical bytes return on every init via the lock file. Committing them bloats the repo for zero reproducibility gain.
❌ A — committed binaries are megabytes of regret
❌ C — state is a different file with different rules
❌ D — /usr/bin is outside the project entirely
🎓 EXAM — Binaries ignored, lock committed — the split never inverts.

Q-D3-03 ✅ **C —** -var wins over everything below it.
🧠 WHY — The ladder resolves top-down at plan time: CLI flags outrank every file, files outrank env vars, everything outranks defaults. The stray flag is always the prime suspect.
❌ A — defaults lose to literally everything
❌ B — env vars sit BELOW files, not above
❌ D — terraform.tfvars loses to auto overlays and CLI
🎓 EXAM — Value surprise? Strip layers top-down: -var, files, env, default.

Q-D3-04 ✅ **D —** workspace show prints only the active name.
🧠 WHY — One word, no metadata — it answers "where am I", not "what exists". Resource lists come from state list; backends from config.
❌ A — regions live in provider blocks
❌ B — list shows names; show shows ONE name
❌ C — buckets live in backend blocks
🎓 EXAM — show = where am I. list = what exists. Never confuse them.

✅ D3-05 · D3-06 · D3-07

FULL WHY

Q-D3-05 ✅ **A —** default survives every delete attempt — it is implicit and permanent.
🧠 WHY — Every backend has a default workspace whether you asked for one or not; deleting it would strand unscoped state. Migrate away from it, never remove it.
❌ B/C/D — named workspaces delete freely once empty — only default is special
🎓 EXAM — Delete-default questions answer themselves: the implicit one is immortal.

Q-D3-06 ✅ **B —** terraform.workspace returns the active name for lookups.
🧠 WHY — It is a read-only string, most often the key into per-env maps for sizing, CIDRs, and tags. It never lists, versions, or addresses anything.
❌ A — backends come from blocks, not expressions
❌ C — resources come from state list
❌ D — versions come from the lock file
🎓 EXAM — workspace in code = a string key. Nothing more, exactly enough.

Q-D3-07 ✅ **C —** validate checks syntax and consistency with zero cloud contact.
🧠 WHY — It parses configuration, resolves references, and type-checks — all locally. Anything requiring AWS (refresh, plan diffs, costs) is outside its universe.
❌ A — refresh needs credentials and APIs
❌ B — sandboxes don't exist in Terraform
❌ D — formatting is fmt's job
🎓 EXAM — Validate = syntax gate. No credentials, no network, no excuses for skipping it.

✅ D3-08 · D3-09 · D4-01

FULL WHY

Q-D3-08 ✅ **D —** Per-team sizes come from variables, not copies.
🧠 WHY — One variable with per-team values (or per-workspace maps) keeps a single codebase honest. Second repos, console edits, and hardcoded locals all fork the truth.
❌ A — repos multiply drift
❌ B — console edits evaporate at next apply
❌ C — hardcoded locals can't vary per caller
🎓 EXAM — Vary by input, never by copy. That sentence ends whole interview threads.

Q-D3-09 ✅ **A —** ~> 5.0 means 5.x automatically, 6.0 never.
🧠 WHY — Pessimistic pinning takes compatible fixes and refuses breaking majors: automation without 3 AM surprises. Bare >= invites the major; exact pins toil on every patch.
❌ B — uncapped ranges install breaking majors silently
❌ C — exact pins freeze bug-fixes too
❌ D — no key floats every release
🎓 EXAM — ~> is the house default. Quote it by expansion: minor-flex, major-freeze.

Q-D4-01 ✅ **A —** terraform fmt -check fails the build on unformatted HCL.
🧠 WHY — Exit 1 on any unformatted file, zero writes — the perfect CI gate. Bare fmt rewrites (for laptops), validate checks syntax (different job), plan previews infra (different universe).
❌ B — rewrites locally — gates must not mutate
❌ C — syntax gate, not style gate
❌ D — previews infrastructure, says nothing about style
🎓 EXAM — Gate = -check. Fix = bare fmt. Different verbs, different jobs.

✅ D4-02 · D4-03 · D4-04

FULL WHY

Q-D4-02 ✅ **A —** Provider tokens live in credentials.tfrc.json.
🧠 WHY — TFC/TFE API tokens are CLI credentials, not configuration: a home-directory JSON block completely outside the repo. Any repo file (tf, tfstate, lock) would leak secrets to every reader.
❌ B — tfstate holds secrets it shouldn't — never ADD more
❌ C — lock holds hashes, not tokens
❌ D — env vars are the ALTERNATE path, not this file
🎓 EXAM — Token + CLI + file → credentials.tfrc.json. Token + CLI + shell → env vars.

Q-D4-03 ✅ **D —** .terraform.lock.hcl pins EXACT builds for reproducibility.
🧠 WHY — Hashes per platform mean every engineer and CI downloads byte-identical binaries. versions.tf declares ranges; the lock freezes them.
❌ A — ranges declare INTENT, the lock freezes REALITY
❌ B — tfstate versions infra, not plugins
❌ C — outputs export values, pin nothing
🎓 EXAM — Intent (versions.tf) vs frozen reality (lock). The exam lives in that gap.

Q-D4-04 ✅ **B —** echo expression | terraform console tests values without files.
🧠 WHY — Console reads stdin, so piped expressions evaluate with zero project context. Writing a .tf for a one-line check is pure toil.
❌ A — -target runs real changes — absurd for a syntax check
❌ C — refresh mutates state — opposite of safe
❌ D — show reads state — your expression isn't there
🎓 EXAM — One-line check = pipe to console. If you touch a file, you overpaid.

✅ D4-05 · D4-06 · D4-07

FULL WHY

Q-D4-05 ✅ **D —** TRACE drowns the log; re-run minimal first.
🧠 WHY — Full TRACE of a big apply is thousands of lines of noise. Minimum reproduction (single resource, lower verbosity) isolates signal before you spend log-reading hours.
❌ A — downgrading past the bug hides the evidence
❌ B — secrets in logs are a DIFFERENT emergency
❌ C — providers log through the same channel — same noise
🎓 EXAM — Big log + weird bug → shrink the blast radius of the reproduction first.

Q-D4-06 ✅ **B —** Format-on-save plus CI -check covers every angle.
🧠 WHY — Prevention at the keyboard, enforcement at the gate: no human discipline required. Either half alone leaks eventually.
❌ A — save-only leaks via web editors and scripts
❌ C — validate checks syntax, not style
❌ D — pre-commit is a THIRD layer, not the pair
🎓 EXAM — Prevention + enforcement: name both halves or the answer is half.

Q-D4-07 ✅ **C —** Keys leave the repo after one scare: env vars and OIDC.
🧠 WHY — Secrets flow through the environment, never through git history. Vaults are the enterprise answer; the repo is never any answer.
❌ A — hardcoded keys are the incident, not the fix
❌ B — tfstate LEAKS secrets — it never stores them safely
❌ D — lock holds hashes — wrong file, wrong job
🎓 EXAM — Keys + repo = the incident. Keys + env/OIDC = the fix.

✅ D4-08 · D5-01 · D5-02

FULL WHY

Q-D4-08 ✅ **C —** Wrappers must preserve exit codes and plan-file flow.
🧠 WHY — Automation decides (gate, notify, apply) by exit status and saved-plan artifacts. Colors, prompts, and committed .terraform dirs are interactive comforts that pipelines must shed, not keep.
❌ A — committing binaries into automation images is bloat
❌ B — color is the first thing pipes strip
❌ D — prompts hang headless runners forever
🎓 EXAM — Pipeline questions: machine-readable out (codes, files), machine-hostile in (prompts, color).

Q-D5-01 ✅ **D —** Third-party module needs source AND version.
🧠 WHY — Source says WHERE, version says WHICH — pinning without a source is a wish, sourcing without a pin is a gamble. Local paths take no version key because there is no registry to resolve.
❌ A — local paths accept no version argument
❌ B — source alone floats every release
❌ C — git refs pin commits, not registry versions
🎓 EXAM — Registry module = two keys or it is wrong. Count the keys in each option.

Q-D5-02 ✅ **B —** VPC module wiring is source, inputs, outputs.
🧠 WHY — The contract is three parts: where the code lives, what you feed it, what it hands back. A fourth invented part (credentials inside, state inside) always marks the distractor.
❌ A — extra invented fourth elements expose distractors
❌ C — missing any of the three breaks the contract
❌ D — provider config flows DOWN, never as module input
🎓 EXAM — Three-part contract: source, in, out. Options with two or four parts die first.

✅ D5-03 · D5-04 · D5-05

FULL WHY

Q-D5-03 ✅ **C —** Init collision comes from .terraform/modules, never the registry.
🧠 WHY — Installed modules cache locally; two copies of one source collide in that directory. The registry is remote and read-only — it cannot collide with anything.
❌ A — the registry serves; it never installs twice
❌ B — state tracks infra, not module code
❌ D — the lock pins providers, not modules
🎓 EXAM — Collision = two local things, one local path. Remote services are never the answer.

Q-D5-04 ✅ **C —** Root calls children; children never call root.
🧠 WHY — Composition flows one way: the root wires child modules together via their outputs. A child reaching upward would tangle every reuse — the exam tests this directionality relentlessly.
❌ A — upward calls would make reuse impossible
❌ B — siblings communicate only through the root
❌ D — state files mirror the same hierarchy
🎓 EXAM — Draw the arrow: root → child. Any option reversing it is wrong.

Q-D5-05 ✅ **A —** Init installs; apply only runs installed code.
🧠 WHY — Module installation is init's job (download, unpack, record). Apply assumes everything is already on disk — new module code without init is invisible to it.
❌ B — plan reads code; it never fetches it
❌ C — validate parses local files only
❌ D — refresh touches state, not module sources
🎓 EXAM — New module + weird behavior → "did you init?" ends the thread.

✅ D5-06 · D5-07 · D5-08

FULL WHY

Q-D5-06 ✅ **D —** Module inputs arrive through variables; outputs leave through outputs.
🧠 WHY — Access scope is the whole game: only declared outputs cross the module boundary. Functions compute, locals stay private, state files are never an interface.
❌ A — functions transform — they transport nothing
❌ B — locals are module-private by design
❌ C — remote state reads STACKS, not modules
🎓 EXAM — Boundary questions: variables in, outputs out. Everything else stays inside.

Q-D5-07 ✅ **B —** Chef-managed hosts are not module backends.
🧠 WHY — Module sources are code locations (local, registry, git, HTTP, S3) — not configuration-management tools. Chef manages machines; it hosts no Terraform code.
❌ A — S3 URLs serve modules fine
❌ C — git is the canonical private source
❌ D — the registry is the default source
🎓 EXAM — Source = where CODE lives. Tools that manage MACHINES never qualify.

Q-D5-08 ✅ **A —** Child values surface as normal references.
🧠 WHY — module.vpc.vpc_id behaves like any other value once outputs are declared — no special syntax, no data source needed. The boundary is crossed at declaration time, not at use time.
❌ B — data sources read PROVIDERS, not child modules
❌ C — remote state reads STACKS, not modules
❌ D — re-declaring duplicates the boundary
🎓 EXAM — If the output is declared, use it like any value. No ceremony.

✅ D6-01 · D6-02 · D6-03

FULL WHY

Q-D6-01 ✅ **D —** Write comes before init in the canonical workflow.
🧠 WHY — Nothing initializes without files to initialize: write → init → plan → apply. Any option starting elsewhere describes a loop, not the lifecycle.
❌ A — plan needs initialized providers
❌ B — apply needs a plan or code
❌ C — init needs SOMETHING to install for
🎓 EXAM — First word of the workflow is write. Stems asking "before init" answer themselves.

Q-D6-02 ✅ **B —** Empty plan with zero counts means clean success.
🧠 WHY — No matching resources = no work = exit 0 with an empty diff. Missing providers, bad syntax, and dead backends all ERROR loudly instead.
❌ A — credential failures error, never empty-plan
❌ C — syntax failures halt before diffing
❌ D — backend failures halt before refreshing
🎓 EXAM — Empty diff + exit 0 = healthy no-op. Empty diff + error = broken. Read the exit code.

Q-D6-03 ✅ **C —** Preview-then-gate is the PR workflow.
🧠 WHY — Plan on every pull request, human review, gated apply: infrastructure gets the same review discipline as application code. Auto-apply on merge skips the review that justifies the pipeline.
❌ A — merge-gated apply skips human judgment
❌ B — console edits bypass everything
❌ D — apply-without-plan is the anti-pattern
🎓 EXAM — PR + plan + gate: the three words of every workflow question.

✅ D6-04 · D6-05 · D6-06

FULL WHY

Q-D6-04 ✅ **C —** Aborting mid-apply leaves partial real resources.
🧠 WHY — Terraform stops issuing new calls, but finished creates stand — the next plan reconciles them. "All-or-nothing rollback" is the fiction; the plan is the recovery tool.
❌ A — completed API calls don't un-happen
❌ B — state is written incrementally — it never waits
❌ D — retries re-enter through a fresh plan, not a resume
🎓 EXAM — Ctrl-C question? Answer: partial stands, next plan heals. Never promise rollback.

Q-D6-05 ✅ **A —** Saved plans bind reviewed code to the apply.
🧠 WHY — apply plan.out executes EXACTLY the reviewed diff — no drift window, no re-evaluation. Re-planning at apply time re-opens the gap the saved plan closed.
❌ B — re-planning re-opens the drift window
❌ C — auto-approve skips the review, not the binding
❌ D — console edits bypass the pipeline entirely
🎓 EXAM — Reviewed plan + saved file = what you saw is what runs.

Q-D6-06 ✅ **A —** destroy -out saves a teardown preview.
🧠 WHY — The destroy plan serializes like any other: review the death list, then apply the file. -target limits scope (different job); auto-approve skips review (opposite job).
❌ B — -target narrows; -out SERIALIZES
❌ C — auto-approve is the anti-review
❌ D — -refresh-only previews drift, not teardown
🎓 EXAM — -out = serialize any plan, including destroy. One flag, every direction.

✅ D6-07 · D6-08 · D6-09

FULL WHY

Q-D6-07 ✅ **A —** Registry version bumps install via init.
🧠 WHY — New module code enters through init (fresh or -upgrade); apply never fetches. Editing the lock by hand fights the tool that owns it.
❌ B — apply runs code; it never downloads it
❌ C — the lock RECORDS installs, it doesn't perform them
❌ D — hand-editing tool-owned files corrupts them
🎓 EXAM — New code in → init. Every time, no exceptions.

Q-D6-08 ✅ **D —** The apply failed — resources may or may not exist.
🧠 WHY — Exit 1 mid-apply guarantees nothing about the world: inspect state and reality, then let the next plan reconcile. "All rolled back" and "all applied" are both fictions.
❌ A — no rollback exists to complete
❌ B — partial creates stand until reconciled
❌ C — the lock releases; the mess remains
🎓 EXAM — Failed apply → inspect, never assume. The next plan is the recovery tool.

Q-D6-09 ✅ **B —** apply -refresh-only syncs state toward reality deliberately.
🧠 WHY — Explicit reconciliation mode: real-world truth flows into state with intent, unlike incidental refresh during plan. Default plans propose changes; destroy previews teardown; validate checks syntax.
❌ A — default plans propose; they don't deliberately sync
❌ C — destroy previews teardown, not truth
❌ D — validate never touches state
🎓 EXAM — Deliberate sync = refresh-only. Incidental refresh rides along.

FULL WHY

✅ D7-01 · D7-02 · D7-03

Q-D7-01 ✅ **A —** State maps config to reality: the binding record.
🧠 WHY — Plan and apply are both joins against this table — without it Terraform cannot tell managed from foreign, desired from actual. Logs, caches, and binaries do none of this.
❌ B — logs narrate; they bind nothing
❌ C — .terraform caches CODE, not bindings
❌ D — binaries execute; they remember nothing
🎓 EXAM — Mapping question → the JSON binding file. Every time.

Q-D7-02 ✅ **B —** Drift = reality moved; plan detects, apply cures.
🧠 WHY — Hands in the console (or outside automation) desync the triangle: config ≠ state ≠ real. Refresh surfaces it as a diff; apply re-asserts code as truth.
❌ A — code edits are INTENT, not drift
❌ C — lock failures block runs; they move nothing
❌ D — version bumps change providers, not reality
🎓 EXAM — Drift trilogy: console cause, plan detection, apply cure. Quote all three.

Q-D7-03 ✅ **C —** Backend block lives INSIDE terraform {}.
🧠 WHY — Backend is a nested block of the terraform settings block — provider blocks configure APIs, variables take inputs, outputs export values. The exam loves this nesting.
❌ A — providers talk to CLOUDS, not state
❌ B — variables take inputs — any type
❌ D — outputs export — they store nothing
🎓 EXAM — State config nests in terraform {}. Say the nesting, not just the name.

✅ D7-04 · D7-05 · D7-06

FULL WHY

Q-D7-04 ✅ **D —** S3 + DynamoDB: bucket holds state, table holds the lock.
🧠 WHY — Two AWS services, two jobs: versioned encrypted storage plus conditional-write locking. Either half alone is half a backend.
❌ A — local paths have no locking at all
❌ B — git versions code — state needs a lock git lacks
❌ C — the registry serves modules, not locks
🎓 EXAM — Name BOTH halves and each half's job. Half answers get half credit.

Q-D7-05 ✅ **D —** Crashed holder? Confirm death, then force-unlock.
🧠 WHY — Locks are protective, not punitive: verify no live process holds it (stale crash), then release by ID. Deleting tables or restoring backups for a lock is surgery for a splinter.
❌ A — deleting the table destroys the locking for EVERYONE
❌ B — backups restore STATE — the lock isn't state
❌ C — waiting on a dead process waits forever
🎓 EXAM — Lock drill: confirm dead → force-unlock by ID. Two steps, in order.

Q-D7-06 ✅ **A —** state rm stops tracking; the resource stands.
🧠 WHY — Surgery on the binding, not the world: Terraform forgets, the cloud keeps billing. Destroy first when you mean destroy; rm when you mean abandon-to-other-management.
❌ B — nothing is deleted — that's the danger AND the feature
❌ C — rm edits STATE, never configuration
❌ D — providers are untouched by state surgery
🎓 EXAM — rm = forget, not destroy. The cloud keeps the resource AND the bill.

✅ D7-07 · D7-08 · D7-09

FULL WHY

Q-D7-07 ✅ **A —** Rename without rebirth = terraform state mv.
🧠 WHY — The binding moves, the resource never notices: new address, same cloud object, empty plan. Taint/replace destroys; rm abandons; mv renames.
❌ B — taint schedules DESTRUCTION — opposite of the ask
❌ C — rm forgets; the new address would RECREATE
❌ D — apply without mv plans a delete+create pair
🎓 EXAM — Rename = mv. Destroy = taint. Forget = rm. Three verbs, three fates.

Q-D7-08 ✅ **D —** Lock fights mean a second writer — coordinate, don't force.
🧠 WHY — Concurrent applies on one state need human sequencing, not lock-breaking: force-unlock under a live apply corrupts state for both runners.
❌ A — force under a LIVE holder corrupts — the cardinal sin
❌ B — retries collide again without coordination
❌ C — the table is healthy; the PROCESS is the conflict
🎓 EXAM — Live lock + force = corruption. Wait, queue, serialize. Force is for the DEAD only.

Q-D7-09 ✅ **B —** Two states, one name: duplicate resources, split brain.
🧠 WHY — Local files don't see each other — each apply births a parallel world under the same names. Locking exists precisely because local state can't do this check.
❌ A — clean no-op requires ONE shared state
❌ C — locks are a REMOTE feature — locals have none
❌ D — providers happily create whatever they're asked
🎓 EXAM — Two writers + no lock = duplicates. The sentence that justifies every backend.

✅ D7-10 · D7-11 · D7-12

FULL WHY

Q-D7-10 ✅ **C —** Migrate with dual backends declared, then init -migrate-state.
🧠 WHY — Both blocks present lets Terraform copy, verify, and flip the pointer in one guided move. Hand-copying JSON or deleting locals invites split-brain.
❌ A — hand JSON copies skip verification
❌ B — deleting local state strands history
❌ D — apply recreates — migration must move, not rebuild
🎓 EXAM — Migration = declare both → init migrates. Never hand-carry state JSON.

Q-D7-11 ✅ **C —** Snapshot raw state BEFORE surgery, verify after.
🧠 WHY — State mv/rm/import rewrite the binding blind: a versioned backup turns mistakes into restores. Planning after (not before) is what confirms the surgery.
❌ A — planning first tells you nothing about the edit
❌ B — locks guard CONCURRENCY, not your typos
❌ D — backups without verification restore blindly
🎓 EXAM — Surgery rule: snapshot, cut, plan-verify. Skip step one, own step four.

Q-D7-12 ✅ **D —** Versioning + encryption + locking = the backend checklist.
🧠 WHY — Every option alone is necessary and insufficient: versions for history, encryption for secrets, locking for teams. The exam's "BEST" questions demand the complete set.
❌ A/B/C — each names ONE pillar and omits two — partial answers
🎓 EXAM — BEST-backend questions: count the pillars. One pillar = wrong. Three = right.

✅ D7-13 · D7-14 · D7-15

FULL WHY

Q-D7-13 ✅ **B —** state replace-provider rewrites every binding at once.
🧠 WHY — One command re-points all provider addresses in the binding — no per-resource surgery, no state deletion, no recreation. It exists precisely for fork migrations.
❌ A — per-resource addressing doesn't scale to whole configs
❌ C — deleting state strands every binding
❌ D — re-apply recreates what should only be re-pointed
🎓 EXAM — Whole-provider moves = one rewrite command. Per-resource tools are for per-resource jobs.

Q-D7-14 ✅ **B —** taint marks for replace; the NEXT plan shows the pair.
🧠 WHY — Taint writes intent into state; destruction happens at apply, previewed as -/+ in the plan. Dirty reads need refresh, not taint; immediate destruction skips the preview.
❌ A — nothing dies until apply runs
❌ C — taint marks STATE, never configuration
❌ D — imports ADD bindings; taints schedule rebirths
🎓 EXAM — Taint = schedule. Plan = preview (-/+). Apply = execute. Three beats, in order.

Q-D7-15 ✅ **C —** Clean plan, empty state list = apply targeted elsewhere.
🧠 WHY — An empty list with a clean diff means the resources live in ANOTHER state (wrong workspace, wrong backend, -target elsewhere) — not that they don't exist.
❌ A — failed applies ERROR — they don't go quiet
❌ B — console edits show as DRIFT, not absence
❌ D — locks block runs; they hide nothing
🎓 EXAM — Clean + empty = looking in the wrong state. Check workspace and backend first.

✅ D7-16 · D7-17 · D7-18

FULL WHY

Q-D7-16 ✅ **D —** Refresh-only previews drift without touching anything.
🧠 WHY — Read-only reconciliation: real-world truth flows into a diff you can inspect, adopt, or revert — zero writes to state or cloud. Destructive options for a read-only question are always wrong.
❌ A — reverting destroys evidence before review
❌ B — force-unlock fixes LOCKS, not drift
❌ C — tainting schedules rebirth for a reading job
🎓 EXAM — Drift review = read-only preview first. Hands off the destroy flags.

Q-D7-17 ✅ **B —** Sensitive values leak through DISPLAY, not storage.
🧠 WHY — State holds secrets regardless; sensitive=true only masks console and log output. Encryption protects storage (different layer); deletion fantasies ignore reality.
❌ A — nothing encrypts just by flagging
❌ C — outputs still carry the value downstream
❌ D — state keeps the plaintext either way
🎓 EXAM — sensitive = display mask. Encryption = storage guard. Two layers, two tools.

Q-D7-18 ✅ **A —** Ephemeral values never touch state.
🧠 WHY — Write-only arguments and ephemeral blocks stay in memory: plans show "(ephemeral)", providers consume, nothing persists. Every other option stores or leaks.
❌ B — sensitive still STORES — it only masks display
❌ C — outputs persist by definition
❌ D — locals live in state-adjacent config
🎓 EXAM — Must-not-persist → ephemeral. The exam's newest favorite word.

✅ D8-01 · D8-02 · D8-03

FULL WHY

Q-D8-01 ✅ **A —** Variables parametrize; locals compute; outputs export.
🧠 WHY — Three blocks, three jobs: inputs vary per caller, locals derive within a module, outputs cross boundaries. Mixing their jobs is the exam's favorite confusion.
❌ B — locals can't take caller input
❌ C — outputs can't feed the same module
❌ D — resources create — they parametrize nothing
🎓 EXAM — Vary, derive, export: say which job before naming the block.

Q-D8-02 ✅ **B —** Objects group related fields into one typed value.
🧠 WHY — One variable carrying name, size, and tags beats three loose strings drifting apart. Maps alone lose per-field types; splitting multiplies wiring.
❌ A — separate variables drift apart
❌ C — maps allow any shape — objects enforce it
❌ D — locals derive; they don't take input
🎓 EXAM — Related fields travel together: object type, one variable.

Q-D8-03 ✅ **C —** Validation rejects bad input at plan time.
🧠 WHY — Rules on variables fail fast with custom messages — before any API call, before any cost. Documentation comments never enforce; apply-time errors bill first.
❌ A — comments describe; they enforce nothing
❌ B — post-apply errors already cost money
❌ D — sensitive masks display — validates nothing
🎓 EXAM — Reject early = validation block. The exam rewards fail-fast everywhere.

✅ D8-04 · D8-05 · D8-06

FULL WHY

Q-D8-04 ✅ **D —** One map + for_each beats N nearly-identical blocks.
🧠 WHY — Data varies, logic stays single: add a map entry, get a subnet. Count-indexed copies and duplicated blocks fork the logic with every edit.
❌ A — duplicated blocks fork on every future edit
❌ B — count breaks on middle removals
❌ C — workspaces isolate ENVS, not subnets
🎓 EXAM — Repeated blocks with varying data = for_each over a map. Reflex-level.

Q-D8-05 ✅ **A —** Dynamic blocks generate repeated nested blocks.
🧠 WHY — Ingress rules, tags, and lifecycle stanzas repeat INSIDE a resource — dynamic + for_each stamps them from data. Splat READS collections; count clones WHOLE resources.
❌ B — splat reads; it never generates
❌ C — count clones resources, not nested blocks
❌ D — workspaces isolate environments
🎓 EXAM — Nested repetition = dynamic. Whole-resource repetition = count/for_each.

Q-D8-06 ✅ **A —** N ingress blocks generate from a list variable.
🧠 WHY — One dynamic block over a list of rule objects: data in, stanzas out. count on the RESOURCE would clone whole security groups — wrong layer entirely.
❌ B — resource-level count clones the whole SG
❌ C — hardcoded rules can't vary per caller
❌ D — outputs export — they generate nothing
🎓 EXAM — Check the layer: nested block → dynamic. Whole resource → count/for_each.

✅ D8-07 · D8-08 · D8-09

FULL WHY

Q-D8-07 ✅ **C —** Data sources READ; they never create.
🧠 WHY — AMIs, existing VPCs, current regions: data blocks query and expose, adding zero resources to the plan. Managed resources create; outputs export; variables input.
❌ A — resources CREATE — the opposite job
❌ B — outputs export values downstream
❌ D — variables carry input, query nothing
🎓 EXAM — Read-only lookup = data source. "Latest AMI" is the tell.

Q-D8-08 ✅ **B —** Orphaned t2.micro + -target: the classic scoped-apply scar.
🧠 WHY — Targeted applies skip everything outside scope — un-targeted resources drift from code silently. The fix is a full apply, not more targeting.
❌ A — deleting state hides the symptom
❌ C — more -target narrows further — wrong direction
❌ D — locks guard concurrency, not scope
🎓 EXAM — -target scars: untargeted resources drift silently. Full apply heals.

Q-D8-09 ✅ **D —** for_each keys must be stable and unique.
🧠 WHY — Keys are identity: renames rebirth, duplicates collide, indices shift. Stable natural keys (names, AZs) survive edits that destroy count-based addressing.
❌ A — indices shift on middle removal — the count trap
❌ B — duplicates collide at plan time
❌ C — renames are destroy+create pairs
🎓 EXAM — Keys = identity. Stable, unique, rename-proof. Three adjectives or wrong.

✅ D8-10 · D8-11 · D8-12

FULL WHY

Q-D8-10 ✅ **B —** Count removal shifts every later index.
🧠 WHY — Index is identity under count: deleting [1] renames [2]→[1], forcing rebirth down the line. for_each keys don't shift — the exam's core repetition lesson.
❌ A — nothing is free — later resources rebirth
❌ C — first-element removal shifts EVERYTHING
❌ D — state edits can't fix index identity
🎓 EXAM — Middle-out under count = cascade. The sentence that sells for_each.

Q-D8-11 ✅ **C —** Splatting collects one attribute across instances.
🧠 WHY — aws_instance.web[*].public_ip projects the fleet into a list: outputs, security rules, and inventories all consume it. for_each BUILDS; dynamic GENERATES; depends_on ORDERS.
❌ A — for_each builds resources, not lists
❌ B — dynamic generates blocks
❌ D — depends_on orders creation
🎓 EXAM — Collect-across = splat. Build-many = for_each. Generate-nested = dynamic.

Q-D8-12 ✅ **D —** Unknown values defer: plan shows (known after apply).
🧠 WHY — Anything derived from not-yet-created resources is unknowable until apply: Terraform marks it, plans around it, resolves later. Errors, nulls, and skips would all lie.
❌ A — unknowns aren't errors — they're futures
❌ B — null would corrupt downstream joins
❌ C — the plan continues around the unknown
🎓 EXAM — (known after apply) = patience, not failure. Read it as a promise.

✅ D8-13 · D8-14 · D8-15

FULL WHY

Q-D8-13 ✅ **A —** 5 subnets from a /16 need newbits = 3 (8 slots).
🧠 WHY — 2^3 = 8 ≥ 5; newbits 2 gives only 4. Round UP to the next power of two.
❌ B — 4 slots can't hold 5 subnets
❌ C — newbits counts BITS, not subnets
❌ D — 8 newbits carves 256 /24s — overkill
🎓 EXAM — Slots = 2^newbits ≥ needed. Round up, always.

Q-D8-14 ✅ **B —** merge() collisions: right side wins.
🧠 WHY — Later maps override earlier ones — the layering primitive. Errors, left-wins, and both-kept all misremember.
❌ A — no error — deterministic override
❌ C — left loses by design
❌ D — one key holds one value
🎓 EXAM — Layering order = argument order. Last writer wins.

Q-D8-15 ✅ **C —** try() guards optional nested access without crashing.
🧠 WHY — First-success-wins with fallback: missing paths return the default instead of erroring. Encryption, speed, and deletion are wrong universes.
❌ A — no encryption anywhere
❌ B — errors become fallbacks — not speed
❌ D — resources are untouched by reads
🎓 EXAM — Optional access = try(). Missing path = fallback, not failure.

✅ D9-02 · D9-03 · D9-04

FULL WHY

Q-D9-02 ✅ **B —** Workspaces isolate STATE, never code or credentials.
🧠 WHY — Same code, N state files: dev and prod differ only in their bindings. Separate repos fork code; separate backends fork truth; shared credentials fork risk.
❌ A — code stays single — that's the point
❌ C — backends stay shared — states differ inside
❌ D — credentials scope separately
🎓 EXAM — Workspace = state fork. Everything else stays shared. One sentence.

Q-D9-03 ✅ **C —** State surgery verbs: mv renames, rm forgets, import adopts.
🧠 WHY — Three verbs, three fates: rename the binding, drop the binding, create a binding. Apply creates RESOURCES — never bindings.
❌ A — apply manages resources, not bindings
❌ B — init installs code, touches no bindings
❌ D — refresh reads reality, rewrites nothing
🎓 EXAM — Binding verbs: mv/rm/import. Resource verbs: plan/apply. Never cross them.

Q-D9-04 ✅ **D —** Splat over for_each instances feeds module outputs.
🧠 WHY — module.instances[*].private_ip collects across keyed instances into the app module's input: collect-then-pass. Count-indexing keyed resources fails; locals can't see inside modules.
❌ A — string keys don't index numerically
❌ B — locals can't pierce module boundaries
❌ C — remote state reads STACKS, not sibling modules
🎓 EXAM — Collect with splat, pass as input. The composition one-liner.

✅ D9-05 · D9-06 · D9-07

FULL WHY

Q-D9-05 ✅ **A —** HCP Terraform runs: remote execution, shared state, governance.
🧠 WHY — The value stack is runs + collaboration + policy: speculative plans on PRs, locked shared state, Sentinel guardrails. Local runs with a remote backend get storage without any of the platform.
❌ B — storage without runs is just a backend
❌ C — local runs keep local blast radius
❌ D — CLI workspaces isolate state — nothing more
🎓 EXAM — Platform = runs + sharing + policy. Backend = storage. Price the difference.

Q-D9-06 ✅ **B —** Speculative plans preview PRs with zero state risk.
🧠 WHY — Dry-run diffs on every pull request: reviewers see infra impact before merge, state untouched, nothing applied. Gates without previews approve blind.
❌ A — blind auto-apply is the anti-pattern
❌ C — locks serialize — they preview nothing
❌ D — local plans can't comment on PRs
🎓 EXAM — PR + preview + no-touch = speculative. The governance flywheel.

Q-D9-07 ✅ **C —** Sentinel GOVERNS with policy-as-code.
🧠 WHY — Mandatory guardrails evaluated at plan time: regions, sizes, tags — violations fail the run before apply. Cost tools estimate; workspaces isolate; runs execute.
❌ A — cost estimation PRICES — Sentinel PERMITS
❌ B — workspaces isolate state, enforce nothing
❌ D — runs execute; policy judges
🎓 EXAM — Sentinel = policy judge. Cost tool = price tag. Different sentences.

✅ D8-16 · D9-08 · D9-01

FULL WHY

Q-D8-16 ✅ **A —** Validation runs at plan, rejecting bad input with your message.
🧠 WHY — Rules on variables fail fast with custom errors — before any API call, before any cost. Decorative-never and console-only misunderstand the feature; after-apply is already too late.
❌ B — decorative-never denies the feature exists
❌ C — console-only shrinks a plan-time gate
❌ D — after apply the money is spent
🎓 EXAM — Reject early = validation block. Fail-fast wins.

Q-D9-08 ✅ **D —** Agents run where your cloud lives; the platform orchestrates.
🧠 WHY — Hybrid split: private network execution, SaaS control plane. All-local loses governance; all-remote can't reach private subnets; workspaces isolate state, not networks.
❌ A — pure local keeps local blast radius
❌ B — pure remote can't touch private nets
❌ C — workspaces isolate STATE, not networks
🎓 EXAM — Private subnets + SaaS control = agents. The hybrid sentence.

Q-D9-01 ✅ **A —** HCP lineage: TFC managed, TFE self-hosted.
🧠 WHY — Same platform, different operators: SaaS vs your basement. The exam tests who patches, who scales, who holds the keys.
❌ B — self-hosted is TFE's definition
❌ C — agents are a NETWORK feature, not a tier
❌ D — workspaces exist in both
🎓 EXAM — Who operates it? SaaS = TFC, self-hosted = TFE.

MOCK KEY

✅ M1-01 → M1-04

Q-M1-01 ✅ **A —** fmt -check gates CI without rewriting.
🧠 WHY — Gate = -check (exit 1, no writes); bare fmt is the local fixer. Rewriting in CI mutates what it judges.

Q-M1-02 ✅ **B —** State maps configuration to real-world objects.
🧠 WHY — The binding table plan/apply join against. Compile-faster/skip-init/encrypt are all wrong-layer fictions.

Q-M1-03 ✅ **C —** 0/0/0 on re-apply = idempotence.
🧠 WHY — Same code, same world, no-op second run. Syntax, backends, and reviews are different properties.

Q-M1-04 ✅ **D —** VPC ID + CIDR land in state as resource metadata.
🧠 WHY — Apply records what the APIs returned. Re-queried-every-second denies state; versions.tf pins providers.

MOCK KEY

✅ M1-05 → M1-08

Q-M1-05 ✅ **A —** count 3→2 destroys web[2], the tail.
🧠 WHY — Index is identity: the highest index vanishes. Survivors keep addresses; nothing else rebirths.

Q-M1-06 ✅ **B —** default is the undeletable workspace.
🧠 WHY — Implicit and permanent — migrate away, never remove. Named envs delete freely once empty.

Q-M1-07 ✅ **C —** for_each takes a set or map.
🧠 WHY — Keys need stability: lists re-index, numbers/strings key nothing. Bare lists wrap with toset().

Q-M1-08 ✅ **D —** TFC vs TFE = hosting: SaaS vs self-hosted.
🧠 WHY — Same platform, different operator. State format, language, and CLI are identical.

MOCK KEY

✅ M1-09 → M1-12

Q-M1-09 ✅ **A —** No-override tag policy = mandatory.
🧠 WHY — Mandatory hard-blocks; soft-mandatory permits override; advisory logs; disabled enforces nothing.

Q-M1-10 ✅ **B —** Login tokens live in credentials.tfrc.json.
🧠 WHY — Home-dir CLI credentials, outside any repo. tfstate/lock/S3 are all wrong-layer or leak-prone.

Q-M1-11 ✅ **C —** Cross-region block needs provider = aws.eu.
🧠 WHY — Alias on the block selects the non-default config. Second installs, depends_on, and second states are all wrong tools.

Q-M1-12 ✅ **D —** TF_VAR vs file vs -var: -var wins.
🧠 WHY — CLI flags outrank every file; files outrank env; env outranks defaults. Strip top-down.

MOCK KEY

✅ M1-13 → M1-16

Q-M1-13 ✅ **A —** Deleted state + apply = duplicate resources.
🧠 WHY — Terraform forgot everything and recreates the world. No auto-restore, no no-op, no speedup.

Q-M1-14 ✅ **B —** Single-cloud deepest-native = CloudFormation.
🧠 WHY — Depth is CFN's whole proposition. Terraform = breadth, Ansible = fleet config, Packer = images.

Q-M1-15 ✅ **C —** Registry modules pin version = "x.y.z".
🧠 WHY — Floating latest breaks future runs; branches aren't versions. Pin every registry source.

Q-M1-16 ✅ **D —** PR → auto plan posted = VCS-driven runs.
🧠 WHY — Remote runs on VCS events with PR comments. Console mode, local-exec, and state mv are unrelated.

MOCK KEY

✅ M1-17 → M1-20

Q-M1-17 ✅ **A —** Users+Nginx on live VMs = Ansible.
🧠 WHY — Fleet configuration of existing hosts is the config-management layer. Provisioners create; they don't babysit.

Q-M1-18 ✅ **C —** Lock error → read the lock info first.
🧠 WHY — Who, what op, since when — then decide stale vs live. Deleting rows and destroying everything are panic moves.

Q-M1-19 ✅ **B —** Workflow order: write, init, plan, apply.
🧠 WHY — Files before init, init before plan, plan before apply. Any rotation describes a loop, not the lifecycle.

Q-M1-20 ✅ **D —** String-key addressing implies for_each keys.
🧠 WHY — ["us-east-1a"] is a map/set key, never a count index. Count addresses numerically.

MOCK KEY

✅ M1-21 → M1-24

Q-M1-21 ✅ **A —** Same word, different concept = workspace.
🧠 WHY — CLI workspaces isolate state; TFC workspaces bind config+vars+state+runs. The exam's favorite homonym.

Q-M1-22 ✅ **B —** Possibly-empty list reads via one(x[*].attr).
🧠 WHY — one() returns the single element or null — x[0] crashes on empty. count.index writes, never reads.

Q-M1-23 ✅ **C —** Saved plans bind reviewed == executed.
🧠 WHY — Serializing the exact diff closes the drift window between review and apply. Re-planning re-opens it.

Q-M1-24 ✅ **D —** Declarative = state end-state, tool derives steps.
🧠 WHY — WHAT not HOW: graph orders, providers translate. Scripting calls and manual ordering are imperative.

MOCK KEY

✅ M1-25 → M1-28

Q-M1-25 ✅ **A —** Remote backends: shared truth, locking, secrets off laptops.
🧠 WHY — Collaboration needs all three. No-locking/no-backend claims deny the feature; code is never encrypted.

Q-M1-26 ✅ **B —** PR cost estimates come from remote-run plan analysis.
🧠 WHY — The platform prices the diff. fmt styles, state mv rebinds, consoles click — none price anything.

Q-M1-27 ✅ **C —** References create graph edges ordering creation.
🧠 WHY — Implicit dependencies from interpolation. No cost, no duplicates — just ordering.

Q-M1-28 ✅ **D —** workspace show prints the active name only.
🧠 WHY — One word answering "where am I". Resources come from state list; types from config.

MOCK KEY

✅ M1-29 → M1-32

Q-M1-29 ✅ **A —** Console tagging surfaces as a ~ drift change.
🧠 WHY — Refresh imports reality into the diff; clean applies never hide hand edits. Absence claims deny refresh.

Q-M1-30 ✅ **B —** TFC workspaces bind config, variables, state, run history.
🧠 WHY — A full execution context, not a key suffix. Suffix-only describes CLI workspaces — the homonym trap.

Q-M1-31 ✅ **C —** ~> 5.0 allows 5.x, never 6.0.
🧠 WHY — Minor-flex, major-freeze. Bare >= invites breaking majors; exact pins toil on patches.

Q-M1-32 ✅ **D —** init never creates real infrastructure.
🧠 WHY — Downloads, backend config, lock file — all local. Creation starts at apply.

MOCK KEY

✅ M1-33 → M1-36

Q-M1-33 ✅ **A —** Soft vs hard mandatory = override possibility.
🧠 WHY — Permitted-with-rights vs never. Language, speed, and nothing are non-answers.

Q-M1-34 ✅ **B —** default survives every delete.
🧠 WHY — Implicit and permanent. Named envs delete freely once empty.

Q-M1-35 ✅ **C —** Noisiest TF_LOG level = TRACE.
🧠 WHY — Full RPC detail for provider bugs. OFF silences, INFO summarizes, WARN filters to warnings.

Q-M1-36 ✅ **D —** State secrets are plaintext — restrict access.
🧠 WHY — No default encryption, no hashing, never absent. Bucket policy + IAM + Vault/SSM references.

MOCK KEY

✅ M1-37 → M1-40

Q-M1-37 ✅ **A —** Children receive values via root input variables.
🧠 WHY — The module call's arguments are the only door in. Console edits, hardcoded locals, and surgery bypass the contract.

Q-M1-38 ✅ **B —** Pulumi = general-purpose languages vs HCL.
🧠 WHY — State, planning, and multi-cloud exist on both sides. Language is the axis; the rest are distractors.

Q-M1-39 ✅ **C —** Local/registry/git/HTTP/S3 = module source types.
🧠 WHY — Where code lives. Backends store state, log levels verbosity, workspaces isolate — different lists.

Q-M1-40 ✅ **D —** { for k,i : k => i.public_ip } builds a name→IP map.
🧠 WHY — Object comprehension keyed by name. Lists come from splat; subnets and errors are wrong shapes.

MOCK KEY

✅ M1-41 → M1-44

Q-M1-41 ✅ **A —** validate = syntax + consistency, no cloud.
🧠 WHY — Local gate: parses, resolves, type-checks. Costs, locks, and DNS all need a network validate never touches.

Q-M1-42 ✅ **B —** Golden rule: never hand-edit state; use state commands.
🧠 WHY — Every fix has a command; hand JSON edits corrupt. Commit/email options violate every other rule too.

Q-M1-43 ✅ **C —** TF_VAR_tags maps need single-quoted HCL.
🧠 WHY — Shell-safe quoting around object syntax. Bare braces split, double quotes interpolate, files are another path.

Q-M1-44 ✅ **D —** state pull = raw JSON snapshot before surgery.
🧠 WHY — Versioned backup enabling restore. Applies, formatting, and upgrades are unrelated verbs.

MOCK KEY

✅ M1-45 → M1-48

Q-M1-45 ✅ **A —** Private registry = blessed org modules/providers.
🧠 WHY — Versioned, reviewed, internal. CPUs, free state, and no-init are fantasy features.

Q-M1-46 ✅ **B —** output_changes gates deploys on output deltas.
🧠 WHY — Machine-readable contract diff for pipelines. Formatting, billing, and DNS are other tools' jobs.

Q-M1-47 ✅ **C —** TFC agents run remote operations on your infra.
🧠 WHY — Private-net execution, SaaS control. Laptops, nothing, and fmt-only miss the hybrid point.

Q-M1-48 ✅ **D —** merge() collisions: right side wins.
🧠 WHY — Later arguments override. Errors, left-wins, and drops are all wrong mental models.

MOCK KEY

✅ M1-49 → M1-52

Q-M1-49 ✅ **A —** Ctrl+C leaves partial state; re-apply continues.
🧠 WHY — Finished creates stand recorded; the next plan reconciles. No rollback exists anywhere.

Q-M1-50 ✅ **B —** Validation runs at plan, rejecting bad input.
🧠 WHY — Fail-fast with custom messages before cost. After-apply/console/never all miss the gate.

Q-M1-51 ✅ **C —** -auto-approve belongs in gated pipelines only.
🧠 WHY — Reviewed plan + automation gate. Laptops, everywhere, and nowhere are all wrong scopes.

Q-M1-52 ✅ **D —** workspace new on a taken name: select it instead.
🧠 WHY — Existence error, not corruption. Init, backends, and state are unrelated to naming.

MOCK KEY

✅ M1-53 → M1-57

Q-M1-53 ✅ **A —** VCS integration posts plans back on the PR.
🧠 WHY — Review where code lives. Email, state, and nowhere break the review loop.

Q-M1-54 ✅ **B —** depends_on covers invisible side-effects only.
🧠 WHY — References first, always. Everywhere-clutter and nowhere-purism both break real configs.

Q-M1-55 ✅ **B —** console = REPL for expressions, no cloud.
🧠 WHY — Pipe values, test functions, touch nothing. Backends, deploys, and registries are other verbs.

Q-M1-56 ✅ **D —** force-unlock needs the lock ID after proving death.
🧠 WHY — Confirm no live holder, release by ID, verify health. Nothing-less corrupts; buckets are unrelated.

Q-M1-57 ✅ **A —** Free-tier caps push to paid tier or scoped usage.
🧠 WHY — Scale honestly: sharing tokens, committing state, and ignoring limits all violate other rules.

MOCK KEY

✅ M2-01 → M2-04

Q-M2-01 ✅ **A —** State never compiles providers.
🧠 WHY — EXCEPT-format: mapping, metadata, and plan-cache are real purposes; compiling is the odd one out.

Q-M2-02 ✅ **B —** Max provider logs: TF_LOG=TRACE redirected to file.
🧠 WHY — TRACE is loudest; stderr redirect captures. OFF silences; fmt/styles touch nothing.

Q-M2-03 ✅ **C —** Same model, same env, every time = idempotence via IaC.
🧠 WHY — Repeatability from versioned declaration. Snowflakes, clicks, and untracked change are the disease.

Q-M2-04 ✅ **D —** Heavy teams outgrow free tier: paid or scoped usage.
🧠 WHY — Local-only, committed state, and disabled plans all trade away the platform's value.

MOCK KEY

✅ M2-05 → M2-08

Q-M2-05 ✅ **A —** web["blue"] implies for_each keys.
🧠 WHY — String-key addressing is map/set identity. Count indexes numerically; backends/provisioners are unrelated.

Q-M2-06 ✅ **A —** Pulumi differs: general-purpose languages vs HCL.
🧠 WHY — Both keep state, plan, and span clouds. Language is the axis — the rest are fictions.

Q-M2-07 ✅ **C —** Safe bug-fix pin: version = "~> 2.0".
🧠 WHY — Pessimistic constraint takes patches, refuses majors. Latest-branches and open ranges gamble.

Q-M2-08 ✅ **D —** TF_VAR_count="three" for number = type error at plan.
🧠 WHY — No auto-correction, no silent ignore — the type system fails fast. Destruction claims are panic.

MOCK KEY

✅ M2-09 → M2-12

Q-M2-09 ✅ **B —** Skipping plan risks unreviewed changes executing blind.
🧠 WHY — Plan is the review artifact. Faster-runs and better-state claims skip the safety the question asks about.

Q-M2-10 ✅ **A —** Change without PR = unreviewed, violates peer review.
🧠 WHY — IaC's review discipline is the point. Faster/required/encrypted all miss the governance ask.

Q-M2-11 ✅ **C —** Deleted lock table = race without serialization; restore it.
🧠 WHY — No lock, no mutual exclusion. Faster/encrypt/skip-init all misread the failure.

Q-M2-12 ✅ **D —** output -json piped to jq serves CI/script consumption.
🧠 WHY — Machine-readable exports for automation. Formatting, init, and locking are other verbs.

MOCK KEY

✅ M2-13 → M2-16

Q-M2-13 ✅ **A —** Speculative PR plans come from VCS-driven runs.
🧠 WHY — Remote plans on pull requests, zero state risk. local-exec, fmt, and console are unrelated.

Q-M2-14 ✅ **B —** Red-hour lock: confirm death, force-unlock by ID, prove health.
🧠 WHY — Stale-crash drill in order. Waiting forever stalls; bucket deletes and laptop reboots miss the object.

Q-M2-15 ✅ **C —** Private registry hosts blessed org versions.
🧠 WHY — Reviewed, versioned, internal modules. Avoiding-init and faster-downloads are fantasies.

Q-M2-16 ✅ **D —** values(map)[*].id returns all IDs as a list.
🧠 WHY — Values projection then splat. One-ID, map-shape, and nothing misread the two-step.

MOCK KEY

✅ M2-17 → M2-20

Q-M2-17 ✅ **A —** workspace list shows all, * marks active.
🧠 WHY — Inventory plus position. Current-only, counts, and costs are other commands' jobs.

Q-M2-18 ✅ **B —** Advisory violation logs; the run continues.
🧠 WHY — Informational by design. Blocks belong to mandatory; deletes and bills are fictions.

Q-M2-19 ✅ **C —** lookup() with default returns the fallback on miss.
🧠 WHY — Three-arg safety net for per-region maps. Crashes need two-arg form; deletes never happen.

Q-M2-20 ✅ **D —** Console SG edits surface as ~ changes to reconcile.
🧠 WHY — Refresh imports reality; plan proposes code-as-truth. Invisible-error-new-workspace all deny refresh.

MOCK KEY

✅ M2-21 → M2-24

Q-M2-21 ✅ **A —** apply tfplan guarantees reviewed == executed.
🧠 WHY — The exact serialized diff runs. No-state/no-backend claims deny the file; init speed is unrelated.

Q-M2-22 ✅ **B —** state show displays one resource's attributes.
🧠 WHY — Single-address inspection. Deletes, formats, and workspace creation are other verbs.

Q-M2-23 ✅ **C —** Mandatory policy with missing tags hard-blocks.
🧠 WHY — No override, no warn-and-continue. Advisory warns; deletes target nothing.

Q-M2-24 ✅ **D —** -refresh-only previews and syncs drift explicitly.
🧠 WHY — Read-only reconciliation on demand. Destroy/init/format are different modes entirely.

MOCK KEY

✅ M2-25 → M2-28

Q-M2-25 ✅ **A —** apply -refresh-only writes synced state toward reality.
🧠 WHY — Explicit reconciliation persists. Nothing-written claims deny the sync; backends and trash-plans are unrelated.

Q-M2-26 ✅ **B —** Remote operations run off-laptop with history.
🧠 WHY — Agents execute; the workspace records. No-history, local-only, and no-variables all invert the model.

Q-M2-27 ✅ **C —** New workspaces start with empty state.
🧠 WHY — Apply fills them. Copied-prod, all-resources, and error claims all misunderstand isolation.

Q-M2-28 ✅ **D —** coalesce() returns the first non-null, non-empty value.
🧠 WHY — Fallback chains in order. Always-last, lists, and nothing misread the function.

MOCK KEY

✅ M2-29 → M2-32

Q-M2-29 ✅ **A —** fmt -recursive covers subfolders too.
🧠 WHY — Whole-tree style in one run. Single-file, state, and registry are wrong scopes.

Q-M2-30 ✅ **B —** Module outputs flow upward to root and consumers.
🧠 WHY — Declaration crosses the boundary at use. Downward-only, nowhere, and S3-direct all invert the contract.

Q-M2-31 ✅ **C —** The root module is the entrypoint config calling children.
🧠 WHY — Composition starts here. State files, registries, and lock files are artifacts, not modules.

Q-M2-32 ✅ **D —** -destroy plan mode previews teardown for later apply.
🧠 WHY — Serialized death list, reviewed first. Creation, init, and formatting are other modes.

MOCK KEY

✅ M2-33 → M2-36

Q-M2-33 ✅ **A —** auto.tfvars vs tfvars: layered lexical overlays vs single file.
🧠 WHY — Multi-file layering in name order. Nothing/comments/speed miss the loading mechanic.

Q-M2-34 ✅ **B —** Piped console tests functions and interpolation with no cloud.
🧠 WHY — Stdin evaluation, zero context. Full applies, locks, and billing are heavier verbs.

Q-M2-35 ✅ **C —** Ansible lacks state and plan for infrastructure.
🧠 WHY — No binding table, no dry-run diffs. Newer/faster/pricier are non-answers.

Q-M2-36 ✅ **D —** Dynamic blocks repeat nested blocks from variables.
🧠 WHY — Data in, stanzas out. Providers, workspaces, and backends repeat at other layers.

MOCK KEY

✅ M2-37 → M2-40

Q-M2-37 ✅ **A —** TFC cost estimates appear on PRs from remote plans.
🧠 WHY — Priced diffs where review happens. State, fmt, and never are wrong venues.

Q-M2-38 ✅ **B —** Prod replaces bare apply with apply tfplan after review.
🧠 WHY — Serialized reviewed diff executes. Destroy, rm, and console edits are anti-patterns.

Q-M2-39 ✅ **C —** Soft-mandatory with rights proceeds with permission.
🧠 WHY — Override with authority, audited. Always-blocks, deletes, and ignores misread the mode.

Q-M2-40 ✅ **D —** state list shows every managed address.
🧠 WHY — The full binding inventory. Bills, sources, and nothing-useful deny its job.

MOCK KEY

✅ M2-41 → M2-44

Q-M2-41 ✅ **A —** Missing registry pin risks floating majors breaking runs.
🧠 WHY — Unpinned floats every release. Nothing/instant/state-loss all misprice the risk.

Q-M2-42 ✅ **B —** Committed tfvars secrets: gitignore + example template + secret store.
🧠 WHY — History rewritten, pattern fixed, secrets relocated. Nothing-normal, git-delete, and filename games all fail.

Q-M2-43 ✅ **C —** depends_on belongs on invisible side-effects only.
🧠 WHY — References first, meta-argument second. Everywhere-clutter and nowhere-purism both break.

Q-M2-44 ✅ **D —** Cloudflare CNAME in one apply shows multi-provider power.
🧠 WHY — Two APIs, one graph, one run. Lock-failure/bug/drift all misread composition.

MOCK KEY

✅ M2-45 → M2-48

Q-M2-45 ✅ **B —** validate needs no cloud — syntax only.
🧠 WHY — Local gate, zero credentials. Cloud/backend/state claims invent requirements.

Q-M2-46 ✅ **B —** Backend blocks take literals or -backend-config only.
🧠 WHY — No variables allowed inside backend blocks — partial config plus flags instead. Vars-aren't-broken; S3/speed are unrelated.

Q-M2-47 ✅ **C —** toset() converts lists to stable sets.
🧠 WHY — for_each needs set/map identity. Map-to-string, number-to-bool, and nothing misread the cast.

Q-M2-48 ✅ **D —** show reads state or plan files, human or JSON.
🧠 WHY — Inspection across formats. DNS, futures, and billing are wrong universes.

MOCK KEY

✅ M2-49 → M2-52

Q-M2-49 ✅ **A —** TRACE is loudest: full RPC detail.
🧠 WHY — Deepest debugging verbosity. OFF silences; quieter/errors-only invert the scale.

Q-M2-50 ✅ **B —** Untracked EC2: state list confirms absence, then import.
🧠 WHY — Prove it's foreign before adopting. Destroy-first, rm-rf, and fmt are panic or nonsense.

Q-M2-51 ✅ **C —** TFC run history lives in the workspace.
🧠 WHY — Config plus vars plus state plus runs. Nowhere/local-only/git all deny the platform.

Q-M2-52 ✅ **D —** Child outputs consume as module.NAME.output.
🧠 WHY — Like any value once declared. Email, console-only, and never invent ceremony.

MOCK KEY

✅ M2-53 → M2-57

Q-M2-53 ✅ **D —** Same model, same env = repeatability.
🧠 WHY — Identical outcomes from versioned code. Risk/slowness/cost are non-answers.

Q-M2-54 ✅ **B —** Workspaces isolate state per env under one backend.
🧠 WHY — Same code, N bindings. Providers, binaries, and nothing are un-isolated.

Q-M2-55 ✅ **C —** -out then apply gives reviewed == executed.
🧠 WHY — Serialized exact diff runs. Faster-init/no-state/no-backend all deny the file.

Q-M2-56 ✅ **D —** one() over empty returns null.
🧠 WHY — Single-or-null semantics for conditional resources. Errors, lists, and zeros misread it.

Q-M2-57 ✅ **A —** Modern refresh = plan -refresh-only.
🧠 WHY — Explicit drift preview/sync. Removed/destroy/fmt all misremember history.

MOCK KEY

✅ M3-01 → M3-04

Q-M3-01 ✅ **A —** Build, change, version = Terraform.
🧠 WHY — The three-verb signature. Packers bake, Ansibles configure, gits version code — not infra.

Q-M3-02 ✅ **B —** Private provider registry hosts blessed org providers.
🧠 WHY — Internal mirror of approved builds. Public-only, skip-init, and billing miss the point.

Q-M3-03 ✅ **C —** init -upgrade re-resolves newest allowed providers, rewrites lock.
🧠 WHY — Controlled freshness within constraints. Destroys/formats/workspaces are other verbs.

Q-M3-04 ✅ **D —** Saved destroy plans via plan -destroy -out.
🧠 WHY — Serialized teardown for review. Login/fmt/console are unrelated verbs.

MOCK KEY

✅ M3-05 → M3-08

Q-M3-05 ✅ **A —** Pinned registry source = best practice.
🧠 WHY — Reproducible installs, no floating majors. Optional/forbidden/slow all invert the rule.

Q-M3-06 ✅ **B —** Remote benefits EXCLUDE local-only secrecy.
🧠 WHY — EXCEPT-format: shared truth, locking, and off-laptop secrets are real; local secrecy contradicts sharing.

Q-M3-07 ✅ **C —** toset() needed: for_each takes sets or maps only.
🧠 WHY — Lists re-index; the cast buys stable identity. Happily/numbers/nothing deny the constraint.

Q-M3-08 ✅ **D —** Ghost-lock recovery starts by reading lock info.
🧠 WHY — Who, what, since — before any action. Reboots, table deletes, and destroys are panic order.

MOCK KEY

✅ M3-09 → M3-12

Q-M3-09 ✅ **A —** Credentials belong in env vars or SSO, never git.
🧠 WHY — Environment-scoped secrets. Committed blocks, state outputs, and lock files all leak.

Q-M3-10 ✅ **B —** OS users and packages on live hosts = Ansible.
🧠 WHY — Config-management layer, existing fleet. Provisioners, CFN-single-cloud, and TFC venue all miss.

Q-M3-11 ✅ **C —** Conditional idiom: count = var.on ? 1 : 0.
🧠 WHY — Ternary-gated existence with safe reads. No-conditionals denies HCL; always-on inverts the ask.

Q-M3-12 ✅ **D —** Infra peer review happens via VCS PRs on HCL.
🧠 WHY — Reviewable text, reviewable diffs. Clicks, state edits, and calls bypass review.

MOCK KEY

✅ M3-13 → M3-16

Q-M3-13 ✅ **A —** values(map)[*].id collects all IDs as a list.
🧠 WHY — Values projection then splat. Keys, nothing, and one-string misread the two-step.

Q-M3-14 ✅ **B —** Agents run plans remotely with history.
🧠 WHY — Off-laptop execution, recorded runs. Laptops, never, and fmt-only invert the model.

Q-M3-15 ✅ **C —** Committed explicit credentials risk scraped secrets in minutes.
🧠 WHY — Bots scan public history instantly. Nothing/faster/better-state deny the threat model.

Q-M3-16 ✅ **D —** Stale state reconciles via refresh during plan.
🧠 WHY — Reality re-read into the diff. New workspaces, config deletes, and fmt are wrong repairs.

MOCK KEY

✅ M3-17 → M3-20

Q-M3-17 ✅ **A —** Speculative PR plans come from VCS-driven runs.
🧠 WHY — Remote dry-runs on pull requests. Cron-fmt, local state, and email hooks are unrelated.

Q-M3-18 ✅ **B —** state show vpc reads one resource's attributes.
🧠 WHY — Single-address inspection. Bills, sources, and nothing deny its job.

Q-M3-19 ✅ **C —** merge() collisions: right side wins.
🧠 WHY — Later arguments override. Left-wins, errors, and neither-drops misremember.

Q-M3-20 ✅ **D —** tfvars vs auto.tfvars: single file vs layered lexical overlays.
🧠 WHY — Explicit -var-file vs automatic layering. Nothing/color/speed miss the loading mechanic.

MOCK KEY

✅ M3-21 → M3-24

Q-M3-21 ✅ **A —** Root passes inputs via module call arguments.
🧠 WHY — The only door in. Surgery, consoles, and email bypass the contract.

Q-M3-22 ✅ **B —** Console tag drift appears as ~ change.
🧠 WHY — Refresh imports reality; plan proposes revert. No-output/new-backend/crash deny refresh.

Q-M3-23 ✅ **C —** coalesce() picks the first non-null value.
🧠 WHY — Ordered fallback chain. Last/all/none misread the function.

Q-M3-24 ✅ **D —** Example tfvars commits; real tfvars stays local.
🧠 WHY — Template shared, secrets local. Speed/state/law are non-reasons.

MOCK KEY

✅ M3-25 → M3-28

Q-M3-25 ✅ **A —** Interrupted apply: review partial state, re-apply.
🧠 WHY — Reconciliation through planning. Deletes, new backends, and fmt are panic moves.

Q-M3-26 ✅ **B —** No manual snowflakes = consistency.
🧠 WHY — Identical systems from code. Cost/slowness/risk are non-answers.

Q-M3-27 ✅ **C —** Go binary, no runtime = the Terraform CLI distribution.
🧠 WHY — Single static executable. Provider-only, state-file, and Sentinel readings miss the subject.

Q-M3-28 ✅ **D —** Two green 0/0/0 applies = idempotence.
🧠 WHY — Repeat no-op from unchanged code. Import/drift/taint are other concepts.

MOCK KEY

✅ M3-29 → M3-32

Q-M3-29 ✅ **A —** try() guards optional access without crashing.
🧠 WHY — Fallback on missing paths. Billing, locks, and DNS are wrong universes.

Q-M3-30 ✅ **B —** Graph edges derive from references.
🧠 WHY — Interpolation orders creation. File order, comments, and filenames order nothing.

Q-M3-31 ✅ **C —** console tests expressions with no cloud.
🧠 WHY — REPL evaluation, zero context. Locks, deploys, and bills are heavier verbs.

Q-M3-32 ✅ **D —** TFC workspaces bind config, vars, state, runs.
🧠 WHY — Full execution context. Binaries, suffix-only, and nothing deny the platform.

MOCK KEY

✅ M3-33 → M3-36

Q-M3-33 ✅ **A —** Multi-cloud in one apply: providers are plugins over RPC.
🧠 WHY — One core, many translators, one graph. Giant-binary/no-state/YAML all deny the architecture.

Q-M3-34 ✅ **B —** PR cost control via remote-run estimates.
🧠 WHY — Priced diffs at review time. Fmt, rm, and taint price nothing.

Q-M3-35 ✅ **C —** Splat over instances gives a list of the attribute.
🧠 WHY — Projection across the fleet. Maps, errors, and backends are wrong shapes.

Q-M3-36 ✅ **D —** Ghost lock: first read lock info — who, what, since.
🧠 WHY — Diagnose before acting. Fmt, nothing, and billing skip the evidence.

MOCK KEY

✅ M3-37 → M3-40

Q-M3-37 ✅ **A —** Deep debugging verbosity = TRACE.
🧠 WHY — Full RPC detail. ERROR/WARN filter; OFF silences.

Q-M3-38 ✅ **B —** apply -refresh-only syncs state toward reality.
🧠 WHY — Explicit reconciliation persists. Style, providers, and regions are other verbs.

Q-M3-39 ✅ **C —** Advisory breach logs; the run continues.
🧠 WHY — Informational by design. Blocks, deletes, and bills belong to other modes.

Q-M3-40 ✅ **D —** Partial apply then re-apply continues from recorded state.
🧠 WHY — No rollback exists; reconciliation moves forward. Rollback/fork/wipe deny the model.

MOCK KEY

✅ M3-41 → M3-44

Q-M3-41 ✅ **A —** Golden images with software baked = Packer.
🧠 WHY — Bakes AMIs; Terraform provisions, provisioners configure last-resort. Console and mv build nothing.

Q-M3-42 ✅ **B —** for_each over a list: wrap with toset().
🧠 WHY — Sets give stable identity; lists re-index. Deletes, depends_on, and count-only dodge the cast.

Q-M3-43 ✅ **C —** TF_VAR objects need single-quoted HCL.
🧠 WHY — Shell-safe object syntax. Bare words split; binaries and no-quoting are nonsense.

Q-M3-44 ✅ **D —** Child outputs consume as module.NAME.output.
🧠 WHY — Like any value once declared. Never/console/email invent ceremony.

MOCK KEY

✅ M3-45 → M3-48

Q-M3-45 ✅ **A —** terraform.tfvars holds auto-loaded variable values.
🧠 WHY — Convention-loaded inputs. Backends, locks, and bills are other files' jobs.

Q-M3-46 ✅ **B —** Per-env lock tables buy parallel applies at table cost.
🧠 WHY — Isolated serialization per env. Nothing/no-init/free-state misprice the trade.

Q-M3-47 ✅ **C —** Validation blocks run at plan.
🧠 WHY — Fail-fast input gates. Never/after-destroy/in-git all miss the timing.

Q-M3-48 ✅ **D —** -destroy plan mode previews teardown.
🧠 WHY — Serialized death list for review. Login/creation/fmt are other modes.

MOCK KEY

✅ M3-49 → M3-53

Q-M3-49 ✅ **A —** New workspace state starts empty; apply fills it.
🧠 WHY — Isolation from zero. Prod-data, errors, and forever-locks deny the model.

Q-M3-50 ✅ **B —** TFC workspaces bind config, vars, state, runs.
🧠 WHY — Full execution context. Suffix-only, nothing, and binaries describe other things.

Q-M3-51 ✅ **C —** Login writes tokens to credentials.tfrc.json.
🧠 WHY — Home-dir CLI credentials. State, lock, and tfvars are wrong files.

Q-M3-52 ✅ **D —** Locals derive values to avoid repetition — single source.
🧠 WHY — Computed once, referenced widely. Duplication, slowness, and state-skips invert the purpose.

Q-M3-53 ✅ **A —** terraform output shows exports after apply.
🧠 WHY — Values for humans and pipelines. Billing, locks, and nothing deny its job.

MOCK KEY

✅ M3-54 → M3-57

Q-M3-54 ✅ **B —** Pin registry modules: floating majors break runs.
🧠 WHY — Reproducibility demands pins. Speed, state-needs, and fmt-does are non-reasons.

Q-M3-55 ✅ **C —** Soft-mandatory with rights proceeds when permitted.
🧠 WHY — Override with authority, audited trail. Blocks/deletes/ignores misread the mode.

Q-M3-56 ✅ **D —** -auto-approve belongs in gated pipelines only.
🧠 WHY — Reviewed plan plus automation gate. Laptops, never-anywhere, and always mis-scope it.

Q-M3-57 ✅ **A —** S3 backend state with secrets is plaintext — restrict plus Vault/SSM.
🧠 WHY — No at-rest encryption by Terraform. Encrypted-absent-hashed are all fictions.

SCORE TRACKER

Know Your Number: 40

TRACKER

57 questions · 70% to pass · **40 correct minimum**. Score every attempt below. Trend matters more than any single run: three straight 44+ means book the exam.

Attempt	Mock 1 /57	Mock 2 /57	Mock 3 /57	Weakest domain	Drill page to revisit
1st pass	__ /57	__ /57	__ /57	_____	_____
2nd pass	__ /57	__ /57	__ /57	_____	_____
Final	__ /57	__ /57	__ /57	_____	_____

Domain drill log (90 drills · target 80%+ per domain)

Domain	Drills	Score 1	Score 2	Ready?
D1 IaC concepts	6	__/6	__/6	☐
D2 Purpose	8	__/8	__/8	☐
D3 Basics	9	__/9	__/9	☐
D4 CLI	8	__/8	__/8	☐
D5 Modules	8	__/8	__/8	☐
D6 Workflow	9	__/9	__/9	☐
D7 State	18	__/18	__/18	☐
D8 Config	16	__/16	__/16	☐
D9 Cloud	8	__/8	__/8	☐

READY RULE

Every domain 80%+ AND two mocks at 44+ on different days. One weak domain fails exams — the blueprint weights breadth.

EXAM DAY

The Night Before & The 60 Minutes

FINALE

You carried state, tamed locks, priced plans, and survived 261 questions. The exam is just one more run — with a timer.

🌙 Night before

① Re-read the **time math** — 63s/question, three passes.
② Skim every **drill key** WHY line once.
③ Sleep. Cramming locks costs more than it buys.

⏱ The 60 minutes

① Pass 1: answer everything, flag the unsure.
② Pass 2: return to flags with fresh eyes.
③ Pass 3: review EXCEPT/NOT stems — they flip meanings.
④ Never leave blanks — no penalty for wrong.

🎯 **TRAP RADAR**

EXCEPT/NOT stems · workspace homonyms (CLI vs TFC) · -var precedence tops · sensitive-masks-display vs encryption · mandatory-blocks vs advisory-logs.

📚 **SERIES RESOURCES**

Vol 1 Foundations · Vol 2 Production · Practice Lab · this Command Center — the full Terraform Master path by Eng. Mohamed Magdy's course.

— The End 🎬 — Good luck. Go take the 40.